

AMBA and AXI Connect Example

Temperature–Humidity Sensor Module

[*Reference*] ■

➤ SoC Peripheral RTC Design Project (프로젝트 최종 계획서 제출)

1. Project Exam

1) Exam_Washing_Machine

2) Exam_Microblaze_Washing_Machine_Block_Design

2. led_btn_Exam

3. AXI_My_IP_LED_Exam

4. AXI_My_IP_FND_Exam

5. AXI_RTC_Exam

6. AXI_PWM

7. AXI_LCD

8. Servo_Motor_Exam

9. DHT11_Module_Exam

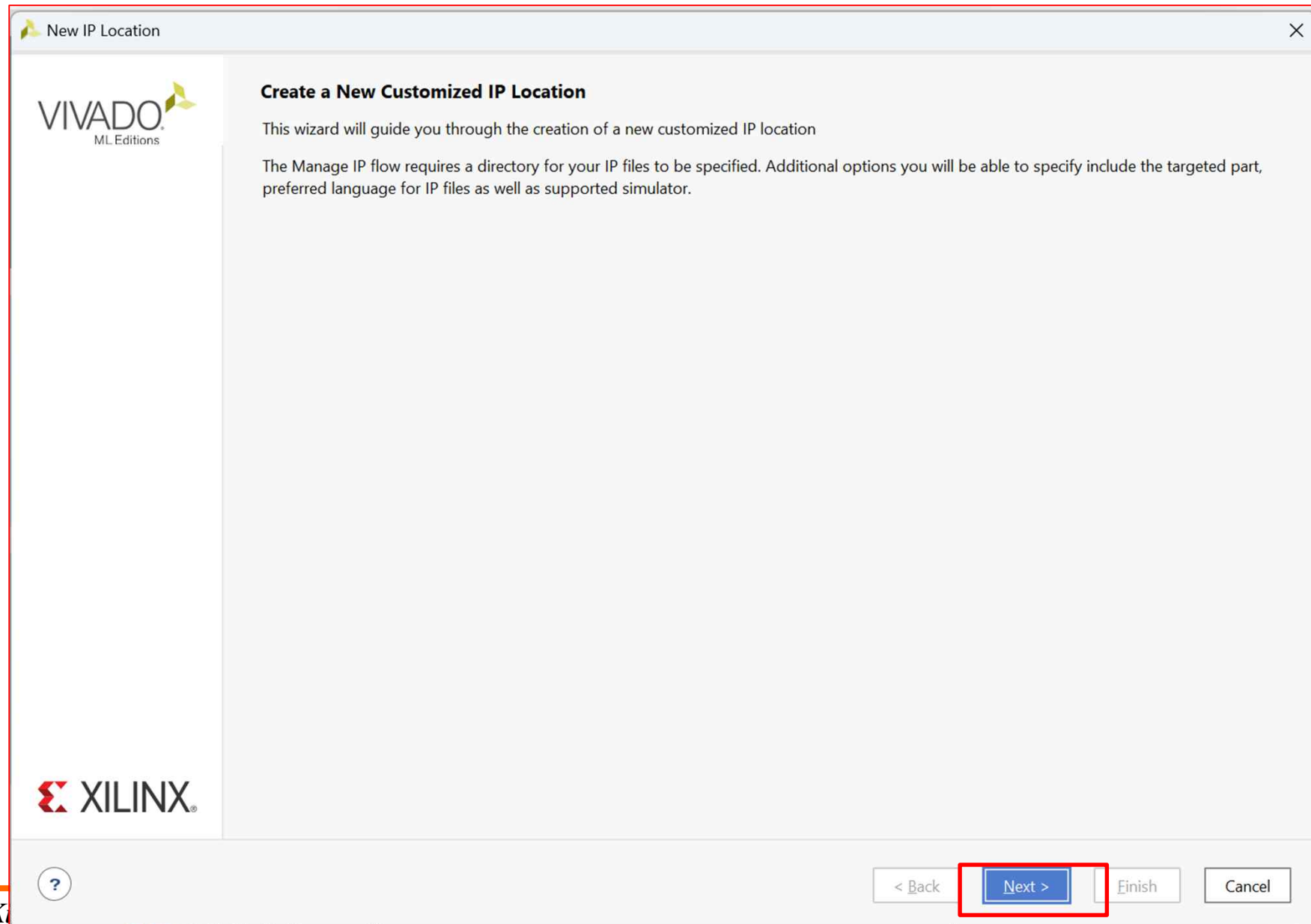
10. Ultrasonic Module Exam

AMBA and AXI Connect Example
Temperature–Humidity Sensor Module
[Create IP]

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

New IP Location

Manage IP Settings
Set options for creating and generating IP.

Part: xc7vx485tffg1157-1 (active) ...

Target language: Verilog

Target simulator: Vivado Simulator

Simulator language: Mixed

IP location: C:/Users/parkj ⓧ ...

? < Back Next > Finish Cancel

➤ AXI Connect ➔ AXI Template

Select Device

Filter, search, and browse parts by their resources. The selection will be applied.

Parts

Boards

To fetch the latest available boards from git repository, click on 'Refresh' button. [Dismiss](#)

Reset [All Filters](#)

Vendor: All Name: All Board Rev: Latest

Search: basys3 (2 matches)

Display Name	Preview	Status	Vendor
Basys3		Download	digilentinc.c
Basys3		Installed	digilentinc.c

Refresh
Catalog was last updated on 09/04/2023 10:24:28 AM

?

OK

Cancel

Basys3 보드 선택

New IP Location

➤ AXI Connect ➔ AXI Template

Manage IP Settings

Set options for creating and generating IP.



Part:	Basys3 (xc7a35tcpg236-1)	...
Target language:	Verilog	▼
Target simulator:	Vivado Simulator	▼
Simulator language:	Mixed	▼
IP location:	C:/Users/parkj	...

원하는 *IP Location* 지정

Finish



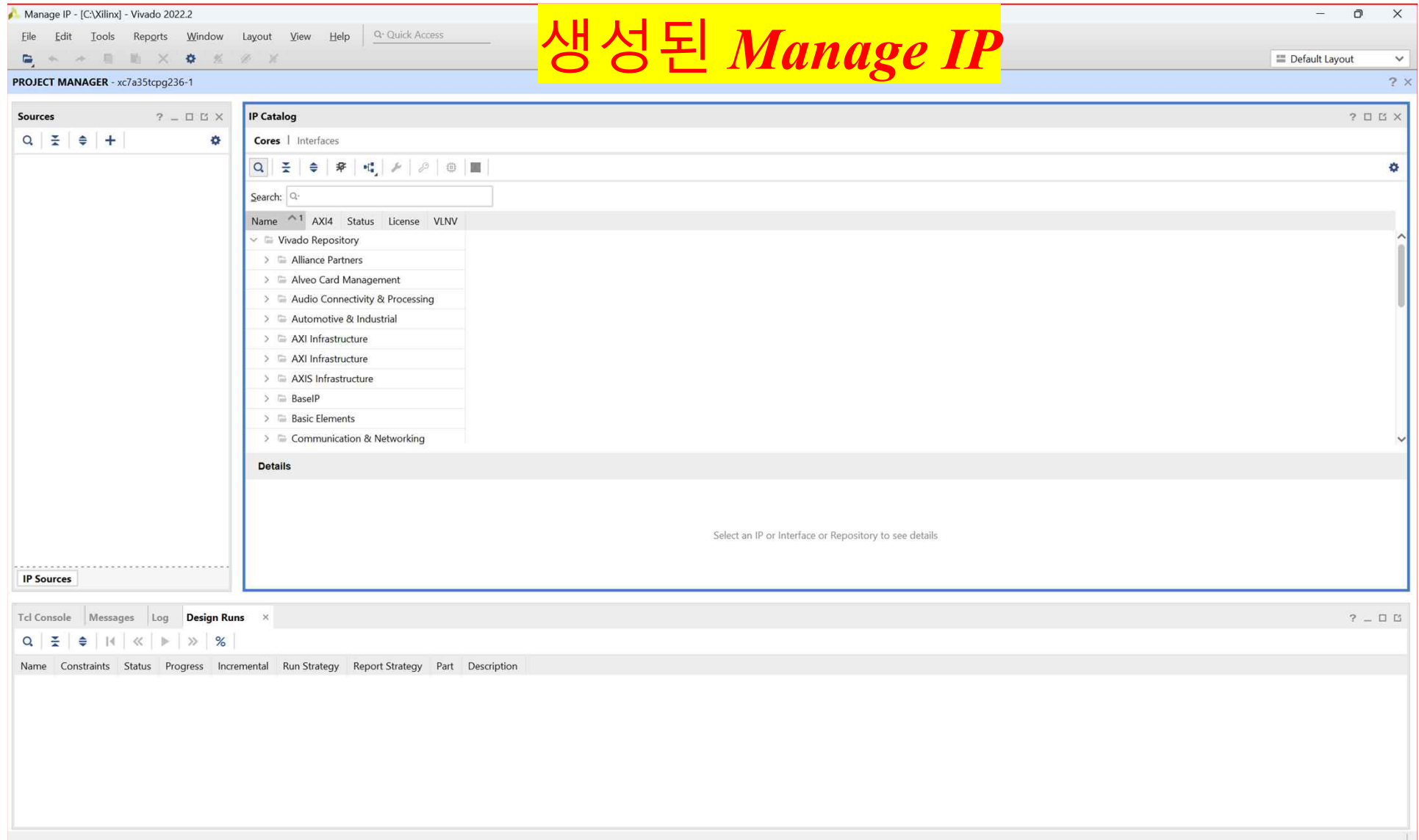
< Back

Next >

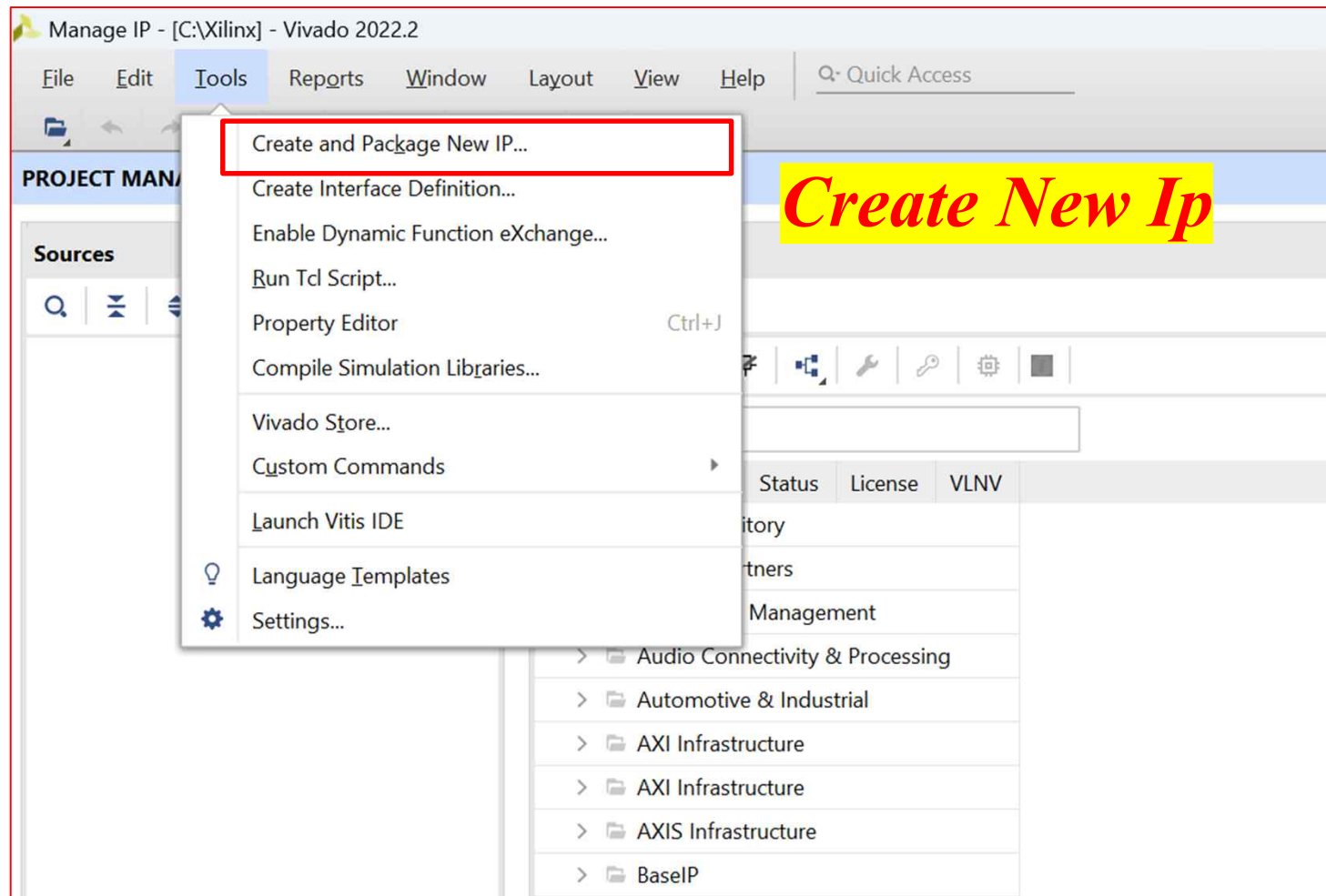
Finish

Cancel

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

Create and Package New IP

Create Peripheral, Package IP or Package a Block Design

Please select one of the following tasks.

Packaging Options

- ☐ Package your current project
Use the project as the source for creating a new IP Definition.
- ☐ Package a block design from the current project
Choose a block design as the source for creating a new IP Definition.
Select a block design: axi_connect_1
- ☐ Package a specified directory
Choose a directory as the source for creating a new IP Definition.

Create AXI4 Peripheral

- ☒ Create a new AXI4 peripheral
Create an AXI4 IP, driver, software test application, IP Integrator AXI4 VIP simulation and debug demonstration design.

Create New Ip

< Back Next > Finish Cancel

➤ AXI Connect ➔ AXI Template

Create and Package New IP

Peripheral Details
Specify name, version and description for the new peripheral

Name: DHT11_axi

Version: 1.0

Display name: DHT11_axi_v1.0

Description: My new AXI IP

IP location: C:/Xilinx/managed_ip_project/./ip_repo

☐ Overwrite existing

이름 변경 후 *Next*

< Back **Next >** Finish Cancel

➤ AXI Connect ➔ AXI Template

Create and Package New IP

Add Interfaces
Add AXI4 interfaces supported by your peripheral

☐ Enable Interrupt Support

Interfaces

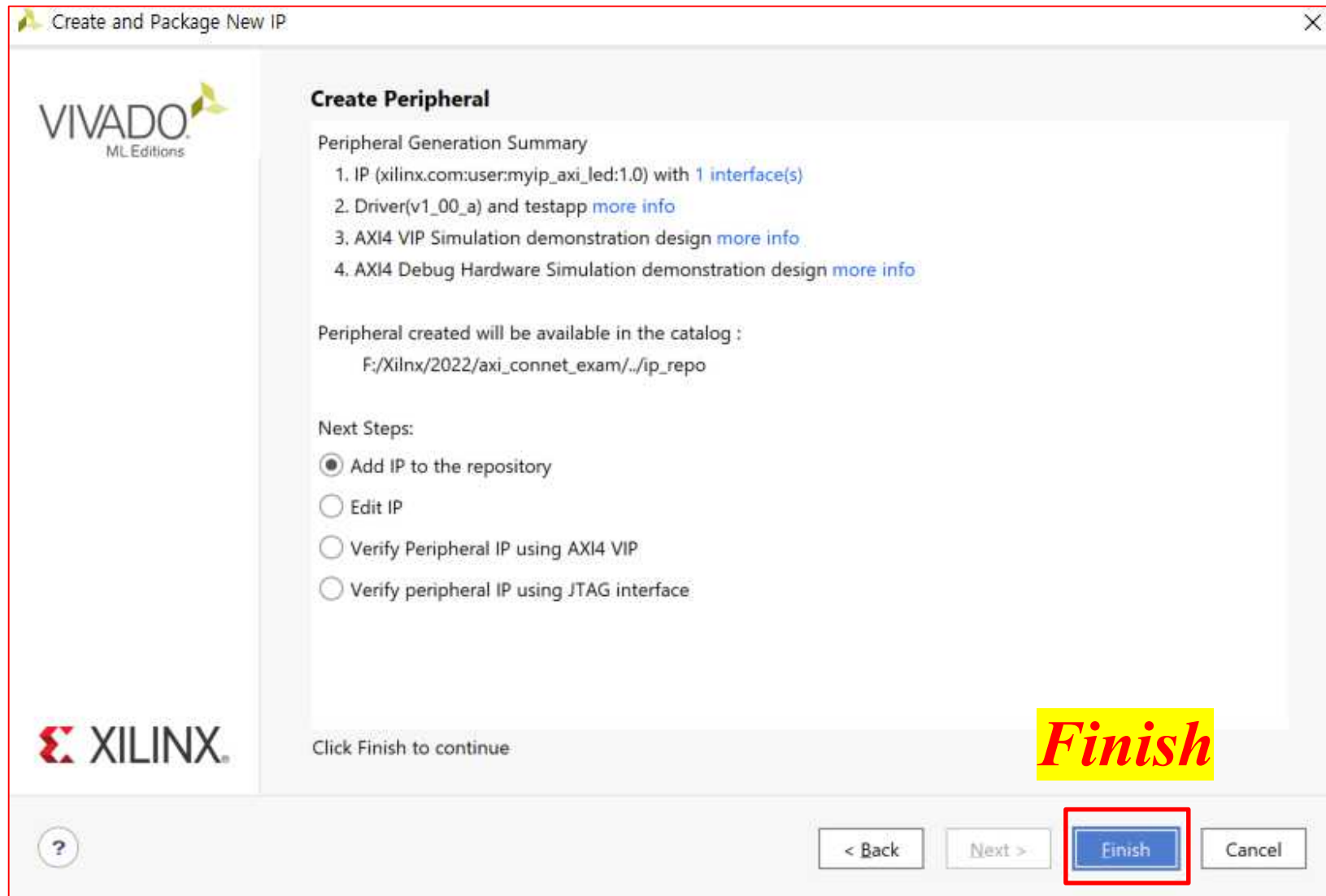
- S00_DHT11_AXI

S00_DHT11_AXI
DHT11_axi_v1.0

Name	S00_DHT11_AXI
Interface Type	Lite
Interface Mode	Slave
Data Width (Bits)	32
Memory Size (Bytes)	64
Number of Registers	4 [4..512]

이름 변경 후 *Next*

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

IP Catalog

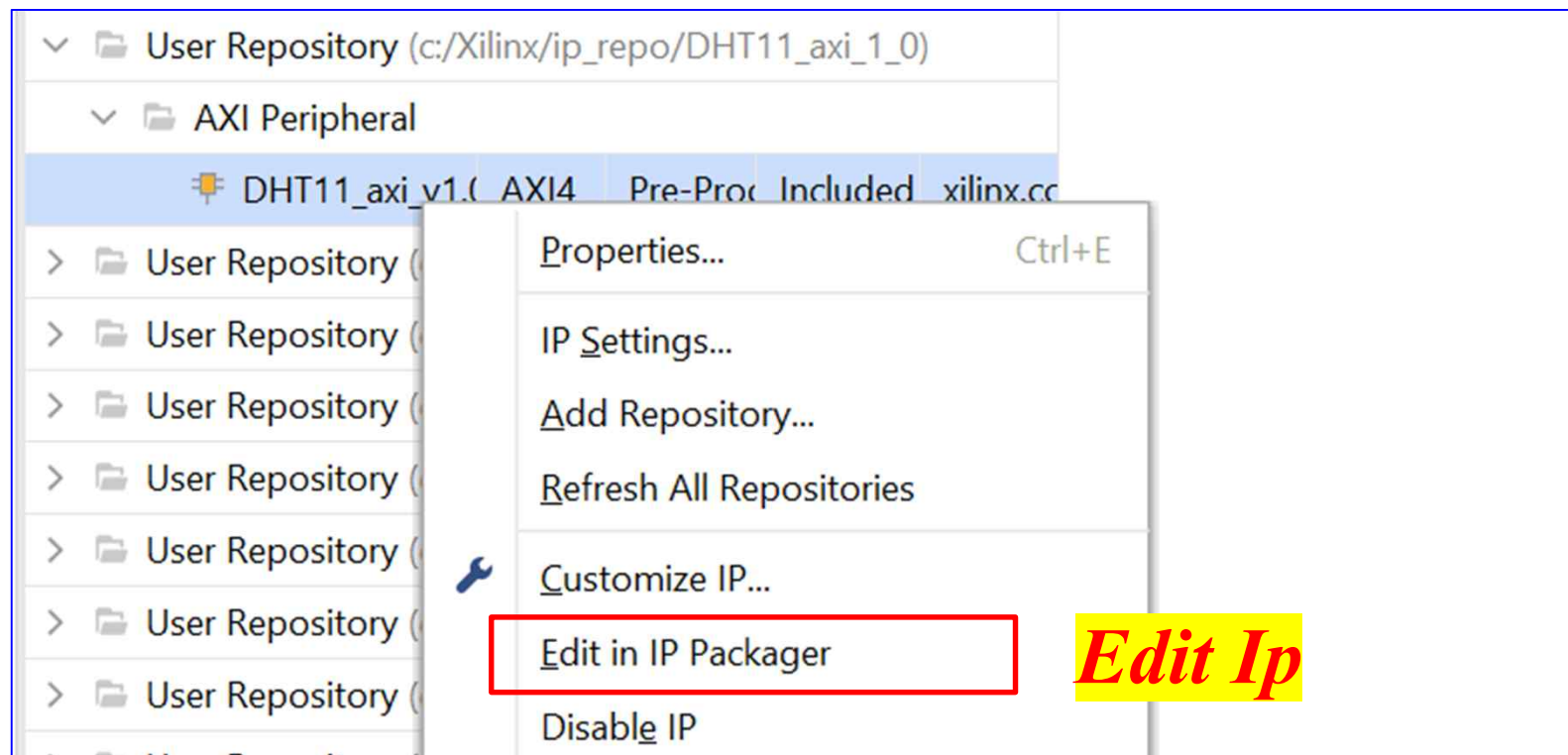
Cores | Interfaces

Search:

Name	AXI4	Status	License	VLNV
- User Repository (c:/Xilinx/ip_repo/DHT11_axi_1_0) AXI Peripheral DHT11_axi_v1.0 AXI4 Pre-Proc Included xilinx.cc - User Repository (c:/Xilinx/ip_repo/ultrasonic_axi_1_0) - User Repository (c:/Xilinx/ip_repo/pwm_axi_1_0) - User Repository (c:/Xilinx/ip_repo/lcd_axi_1_0)				

**IP 생성 확인
(ip_repo에 저장)**

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

The screenshot displays the Xilinx IDE interface with the 'Package IP - DHT11_axi' project open. The 'Sources' pane on the left shows the project hierarchy, with 'DHT11_axi_v1_0 (DHT11_axi_v1_0.v) (1)' highlighted under 'Design Sources (2)'. A red box highlights this entry. Below the 'Sources' pane, a yellow box contains the Korean text '저장된 IP 모듈 확인' (Check saved IP module). The 'Source File Properties' pane at the bottom left shows the properties for 'DHT11_axi_v1_0.v', including its location, type (Verilog), size (2.6 KB), and modification date. The 'Project Summary' pane on the right shows the 'Packaging Steps' and 'Identification' sections. The 'Packaging Steps' section lists various steps with green checkmarks, indicating successful completion. The 'Identification' section displays the IP's metadata, including vendor (xilinx.com), library (user), name (DHT11_axi), version (1.0), display name (DHT11_axi_v1.0), description (My new AXI IP), and root directory.

Sources

- Design Sources (2)
 - DHT11_axi_v1_0 (DHT11_axi_v1_0.v) (1)**
 - IP-XACT (1)
- Constraints
- Simulation Sources (1)
- Utility Sources

Source File Properties

DHT11_axi_v1_0.v

- ☒ Enabled
- Location: c:/Xilinx/ip_repo/DHT11_axi_1_0/hdl
- Type: Verilog
- Library:
- Size: 2.6 KB
- Modified: Today at 11:09:07 AM
- Read-only: No

Project Summary

Packaging Steps

- Identification
- Compatibility
- File Groups
- Customization Parameters
- Ports and Interfaces
- Addressing and Memory
- Customization GUI
- Review and Package

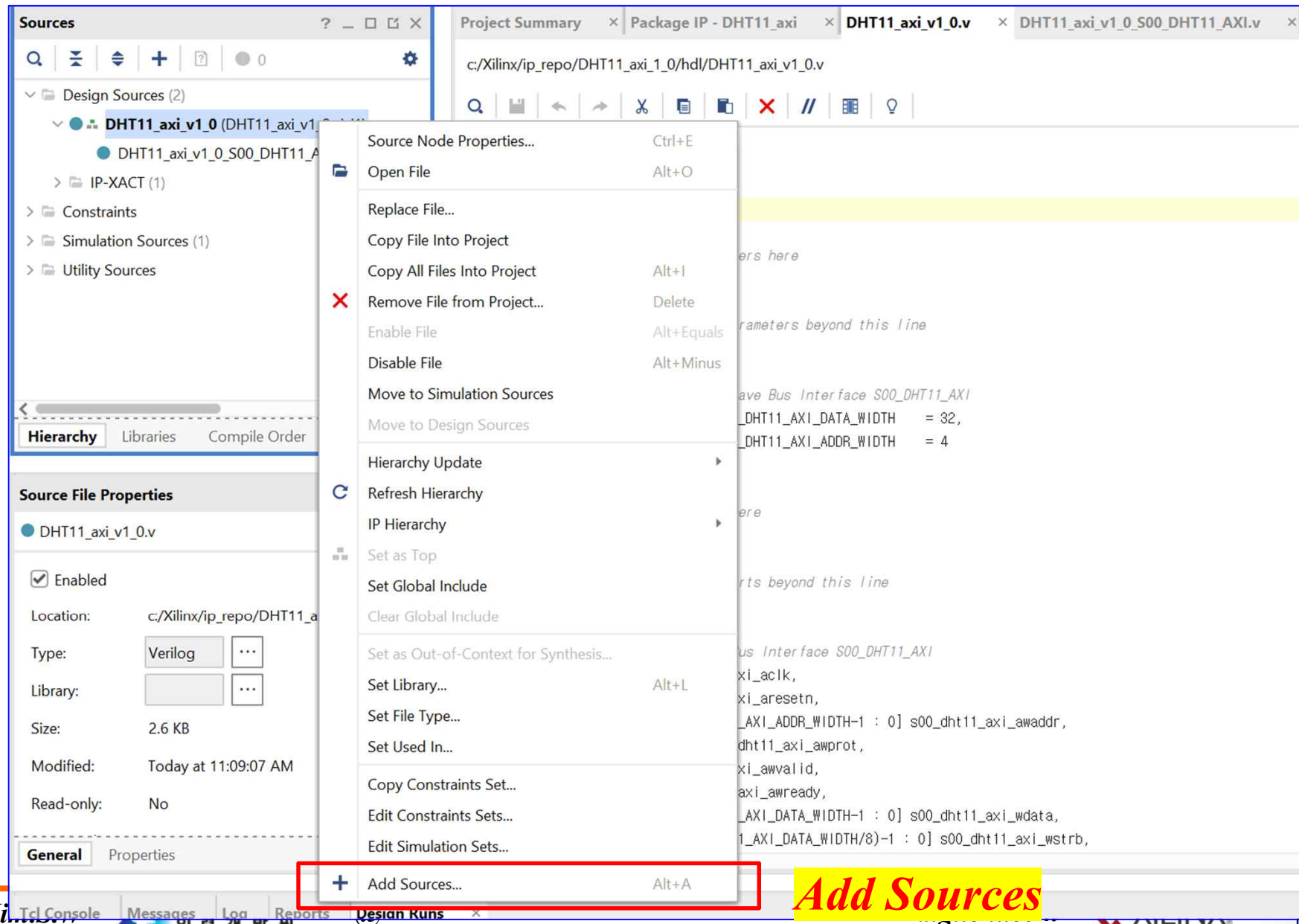
Identification

- Vendor: xilinx.com
- Library: user
- Name: DHT11_axi
- Version: 1.0
- Display name: DHT11_axi_v1.0
- Description: My new AXI IP
- Vendor display name:
- Company url:
- Root directory: c:/Xilinx/ip_repo/DHT11_axi_1_0
- Xml file name: c:/Xilinx/ip_repo/DHT11_axi_1_0/compo

Categories

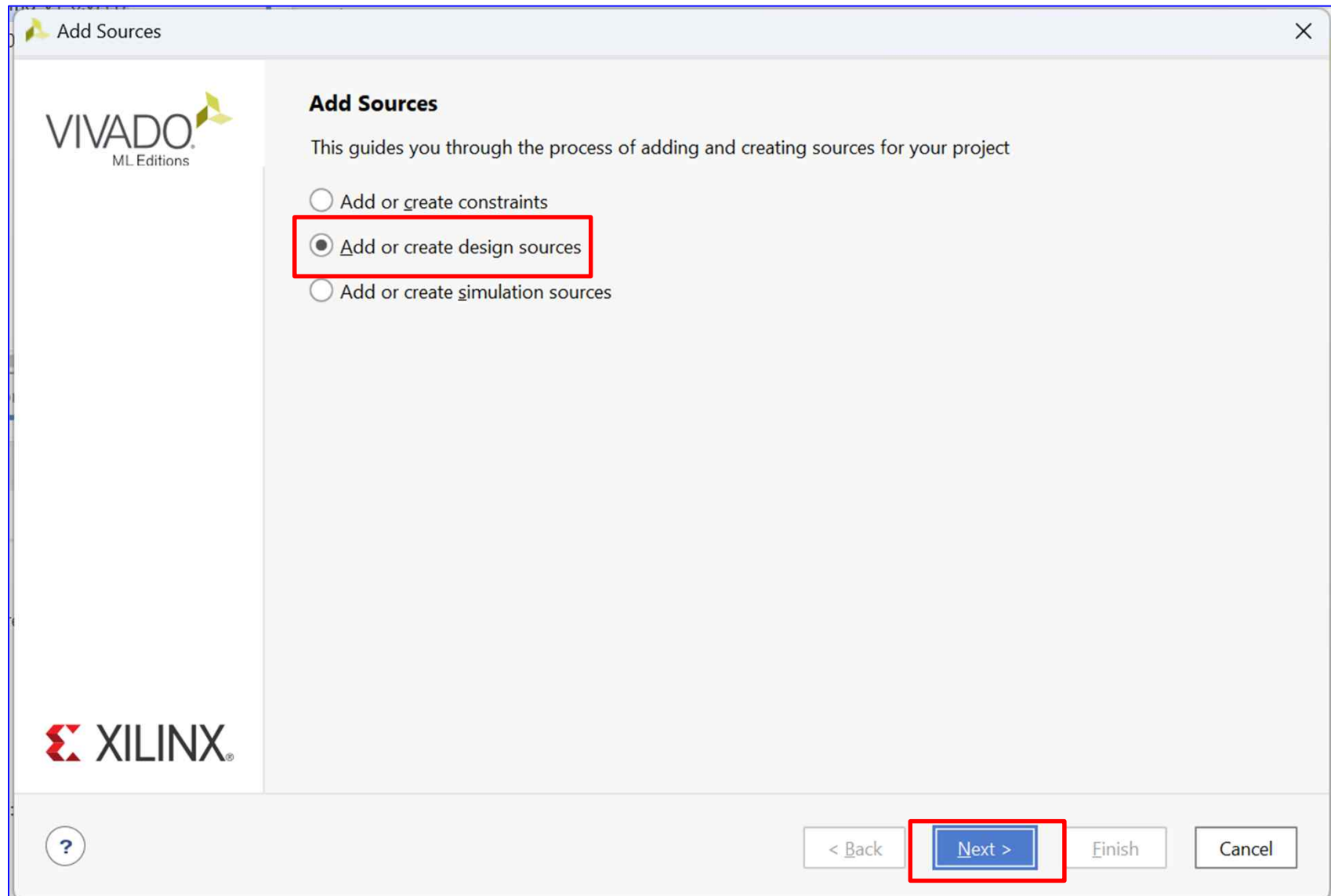
- AXI_Peripheral

➤ AXI Connect ➔ AXI Template

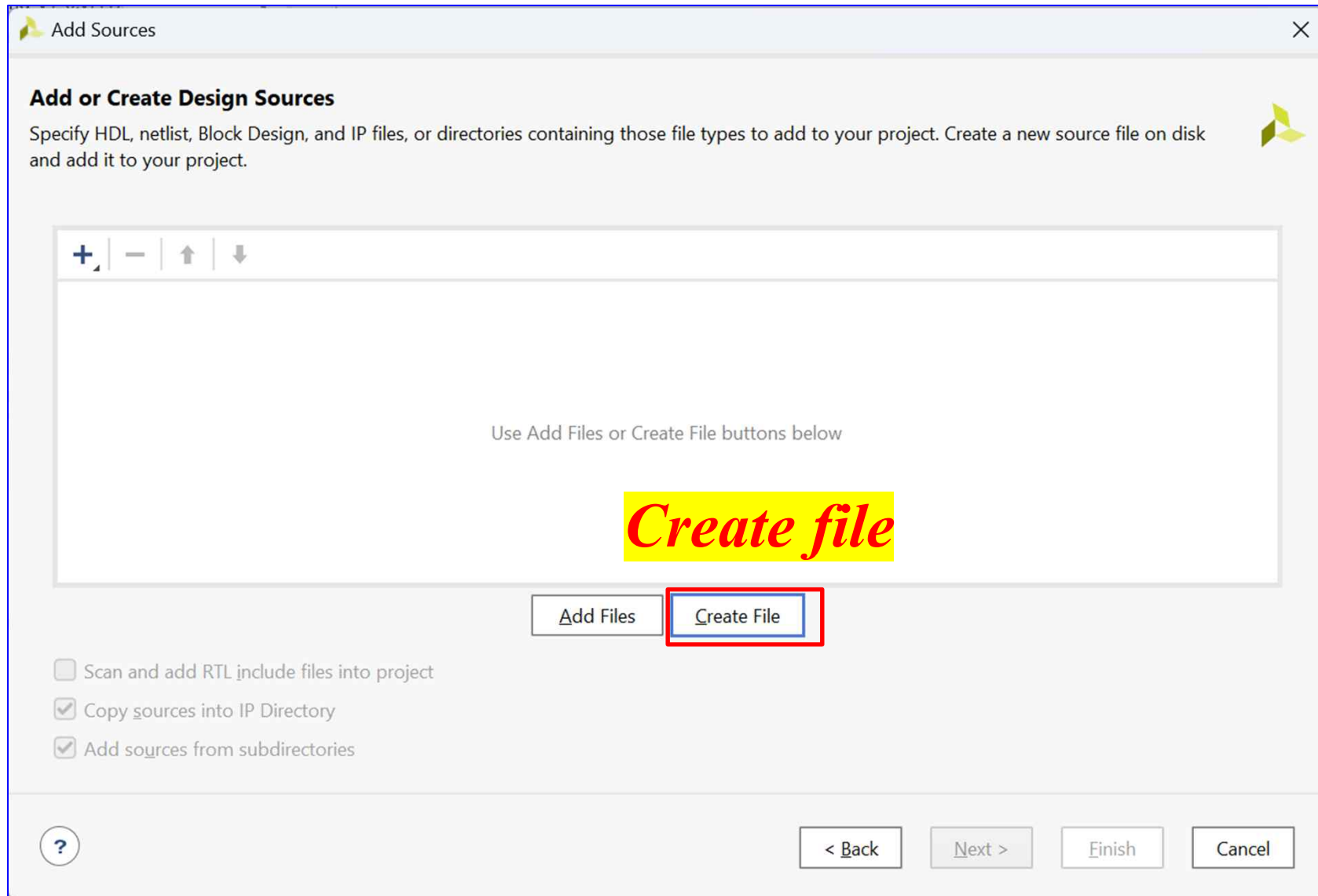


Add Sources

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

Add Sources

Add or Create Design Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those file types to add to your project. Create a new source file on disk and add it to your project.

	Index	Name	Library	Location
	1	DHT11.v	xil_defaultlib	C:/Xilinx
	2	clockdivider_1Mhz.v	xil_defaultlib	C:/Xilinx
	3	edge_detector.v	xil_defaultlib	C:/Xilinx

Add Files
Create File

☐ Scan and add RTL include files into project
☒ Copy sources into IP Directory
☒ Add sources from subdirectories

?
< Back
Next >
Finish
Cancel

Source 추가

➤ AXI Connect ➔ AXI Template

```

1. module DHT11(
2.   input clk, reset, clk1Mhz,
3.   inout dht11_data,
4.   output [3:0] humidity10, humidity0,
5.   output [3:0] temperature10, temperature0
6. );
7.   reg [7:0] humidity, temperature;
8.
9.   assign humidity10 = humidity/10;
10.  assign humidity0 = humidity%10;
11.  assign temperature10 = temperature/10;
12.  assign temperature0 = temperature%10;
13.
14.  parameter S_IDLE = 3'b000;
15.  parameter S_LOW_18MS = 3'b001;
16.  parameter S_HIGH_20US = 3'b010;
17.  parameter S_LOW_80US = 3'b011;
18.
19.  parameter S_HIGH_80US = 3'b100;
20.  parameter S_READ_DATA = 3'b101;
21.  parameter S_WAIT_PEDGE = 3'b110;
22.  parameter S_WAIT_NEDGE = 3'b111;

```

```

23. wire dht_nedge, dht_pedge;
24.   reg count_usec_e;
25.   reg [21:0] count_usec;
26.   reg [2:0] state, next_state, read_state;
27.   reg dht11_data_oe, dht_buffer;
28.   reg [5:0] data_count;
29.   reg [39:0] temp_data;
30.   assign dht11_data = dht_buffer;
31. always @(negedge clk1Mhz or negedge
reset) begin
32.   if(~reset) count_usec = 0;
33.   else begin
34.     if(count_usec_e)
35.       count_usec = count_usec + 1;
36.     else if(!count_usec_e) count_usec = 0;
37.   end
38. end
39. always @(negedge clk1Mhz or negedge
reset)begin
40.   if(~reset)state = S_IDLE;
41.   else state = next_state;
42. end
43.
44. edge_detector ed_dec
(.clk(clk1Mhz), .cp_in(dht11_data), .reset(
reset), .n_edge(dht_nedge), .p_edge(dht_pe
dge));

```

```

45. always @(posedge clk1Mhz or negedge
reset)begin
46.   if(~reset)begin
47.     count_usec_e <= 0;
48.     next_state <= S_IDLE;
49.     dht11_data_oe <= 0;
50.     dht_buffer <= 1'bz;
51.     read_state <= S_WAIT_PEDGE;
52.     data_count <= 0;
53.
54.   end
55.   else begin
56.     case(state)
57.       S_IDLE : begin
58.
59.         if(count_usec < 22'd3_000_000) begin
60.           count_usec_e <= 1;
61.           dht_buffer = 1'bz;
62.         end
63.       else begin
64.
65.         next_state <= S_LOW_18MS;
66.         count_usec_e <= 0;
67.       end
68.     end

```

➤ AXI Connect ➔ AXI Template

```

69. S_LOW_18MS : begin
70.
71.   if(count_usec < 22'd20_000) begin
72.       count_usec_e <= 1;
73.       dht_buffer <= 0;
74.   end
75. else begin
76.     count_usec_e <= 0;
77.     next_state <= S_HIGH_20US;
78.     dht_buffer = 1'bz;
79.   end
80. end

81. S_HIGH_20US : begin
82.
83.   if(count_usec < 22'd70) begin
84.       dht_buffer = 1'bz;
85.       count_usec_e <= 1;
86.       if(dht_nedge)begin
87.           next_state <= S_LOW_80US;
88.           count_usec_e <= 0;
89.       end
90.   else begin next_state <= S_HIGH_20US;
91.       count_usec_e <= 1;
92.   end
93. end
94. else begin
95.     next_state <= S_IDLE;
96.     count_usec_e <= 0;
97.   end
98. end

```

```

99. S_LOW_80US : begin
100.
101.   if(count_usec < 22'd90)begin
102.
103.       if(dht_pedge)begin
104.           next_state <= S_HIGH_80US;
105.           count_usec_e <= 0;
106.       end
107.   else begin
108.       next_state <= S_LOW_80US;
109.       count_usec_e <= 1;
110.   end
111. end else begin
112.     next_state <= S_IDLE;
113.     count_usec_e <= 0;
114. end

115.S_HIGH_80US : begin
116.
117.   if(count_usec < 22'd90)begin
118.
119.       if(dht_nedge)begin
120.           next_state <= S_READ_DATA;
121.           count_usec_e <= 0;
122.       end
123.   else begin
124.       next_state <= S_HIGH_80US;
125.       count_usec_e <= 1;
126.   end
127. end else begin
128.     next_state <= S_IDLE;
129.     count_usec_e <= 0;
130. end
131. end

```

```

132. S_READ_DATA : begin
133.
134.   case(read_state)
135.       S_WAIT_PEDGE: begin
136.           if(dht_pedge) begin
137.               read_state <= S_WAIT_NEDGE;
138.               count_usec_e <= 1;
139.           end
140.       else begin
141.           count_usec_e = 0;
142.       end
143.   end

144. S_WAIT_NEDGE: begin
145.       if(dht_nedge) begin
146.           data_count <= data_count + 1;
147.           read_state <= S_WAIT_PEDGE;
148.       end
149.   else begin
150.       temp_data <= {temp_data[38:0], 1'b0};
151.   end
152.   else begin
153.       temp_data <= {temp_data[38:0], 1'b1};
154.   end
155. end
156. else begin
157.     count_usec_e <= 1;
158.     read_state <= S_WAIT_NEDGE;
159. end
160. default: read_state = S_WAIT_PEDGE;
161. endcase

```

➤ AXI Connect ➔ AXI Template

```
162.if(data_count >= 40)begin
163.
164.  data_count <= 0;
165.  next_state <= S_IDLE;

166. if(temp_data[39:32] +
      temp_data[31:24] + temp_data[23:16]
      + temp_data[15:8] == temp_data[7:0])
      begin

167.  humidity <= temp_data[39:32];
168.  temperature <= temp_data[23:16];
169.  end
170. end
171. end
172. default : next_state = S_IDLE;
173. endcase
174. end
175. end
176.
177.endmodule
```



```

1. `timescale 1ns / 1ps
2. module clockdivider (
3.     input clk,
4.     input reset,
5.     output reg clk1Mhz
6. );
7.     reg [25:0] cnt = 0;
8.     always @(posedge clk or negedge
reset) begin
9.         if (~reset) begin
10.             cnt <= 0;
11.             clk1Mhz <= 0;
12.         end else begin
13.             if (cnt == (50 - 1)) begin
14.                 cnt <= 0;
15.                 clk1Mhz <= ~clk1Mhz;
16.             end else begin
17.                 cnt <= cnt + 1;
18.             end
19.         end
20.     end
21. endmodule

```

```

1. `timescale 1ns / 1ps
2. module edge_detector(
3.     input clk,
4.     input cp_in,
5.     input reset,
6.     output p_edge,
7.     output n_edge
8. );
9.
10.     reg cp_in_old, cp_in_cur;
11.     always @(posedge clk or negedge reset) begin
12.         if (~reset) begin
13.             cp_in_old = 0; cp_in_cur = 0;
14.         end else begin
15.             cp_in_old <= cp_in_cur;
16.             cp_in_cur <= cp_in;
17.         end
18.     end
19.
20.     assign p_edge = ~cp_in_old & cp_in_cur;
21.     assign n_edge = cp_in_old & ~cp_in_cur;
22. endmodule

```

➤ AXI Connect ➔ AXI Template

```

1 timescale 1 ns / 1 ps
2
3 module dht11_v1_0 #
4 (
5     // Users to add parameters here
6
7     // User parameters ends
8     // Do not modify the parameters beyond this line
9
10
11
12     // Parameters of Axi Slave Bus Interface S00_dht11_AXI
13     parameter integer C_S00_dht11_AXI_DATA_WIDTH = 32,
14     parameter integer C_S00_dht11_AXI_ADDR_WIDTH = 4
15 )
16 (
17     // Users to add ports here
18     inout wire dht11_data,
19     // User ports ends
20     // Do not modify the ports beyond this line
21
22
23     // Ports of Axi Slave Bus Interface S00_dht11_AXI

```

```

14 parameter integer C_S_AXI_ADDR_WIDTH = 4
15 )
16 (
17     // Users to add ports here
18     inout wire dht11_data,
19     // User ports ends
20     // Do not modify the ports beyond this line
21
22
23

```

```

46 // Instantiation of Axi Bus Interface S00_dht11_AXI
47 dht11_v1_0_S00_dht11_AXI # (
48     .C_S_AXI_DATA_WIDTH(C_S00_dht11_AXI_DATA_WIDTH),
49     .C_S_AXI_ADDR_WIDTH(C_S00_dht11_AXI_ADDR_WIDTH)
50 ) dht11_v1_0_S00_dht11_AXI_inst (
51     .S_AXI_ACLK(s00_dht11_axi_aclk),
52     .S_AXI_ARESETN(s00_dht11_axi_aresetn),
53     .S_AXI_AWADDR(s00_dht11_axi_awaddr),
54     .S_AXI_AWPROT(s00_dht11_axi_awprot),
55     .S_AXI_AWVALID(s00_dht11_axi_awvalid),
56     .S_AXI_AWREADY(s00_dht11_axi_awready),
57     .S_AXI_WDATA(s00_dht11_axi_wdata),
58     .S_AXI_WSTRB(s00_dht11_axi_wstrb),
59     .S_AXI_WVALID(s00_dht11_axi_wvalid),
60     .S_AXI_WREADY(s00_dht11_axi_wready),
61     .S_AXI_BRESP(s00_dht11_axi_bresp),
62     .S_AXI_BVALID(s00_dht11_axi_bvalid),
63     .S_AXI_BREADY(s00_dht11_axi_bready),
64     .S_AXI_ARADDR(s00_dht11_axi_araddr),
65     .S_AXI_ARPROT(s00_dht11_axi_arprot),
66     .S_AXI_ARVALID(s00_dht11_axi_arvalid),
67     .S_AXI_ARREADY(s00_dht11_axi_arready),
68     .S_AXI_RDATA(s00_dht11_axi_rdata),
69     .S_AXI_RRESP(s00_dht11_axi_rresp),
70     .S_AXI_RVALID(s00_dht11_axi_rvalid),
71     .S_AXI_RREADY(s00_dht11_axi_rready),
72     .dht11_data(dht11_data)
73 );
74
75 // Add user logic here

```

➤ AXI Connect ➔ AXI Template

```

Package IP - dht11  x dht11_v1_0.v  x dht11_v1_0_S00_dht11_AXI.v  x
h:/Xilinx/2023/Xilinx/ip_repo/dht11_1_0/hdl/dht11_v1_0_S00_dht11_AXI.v

371 // Address decoding for reading registers
372 case ( axi_araddr[ADDR_LSB+OPT_MEM_ADDR_BITS:ADDR_LSB] )
373     2'h0 : reg_data_out <= temperature0;
374     2'h1 : reg_data_out <= temperature10;
375     2'h2 : reg_data_out <= humidity0;
376     2'h3 : reg_data_out <= humidity10;
377     default : reg_data_out <= 0;
378 endcase
379 end
380
381 // Output register or memory read data
    
```

변경

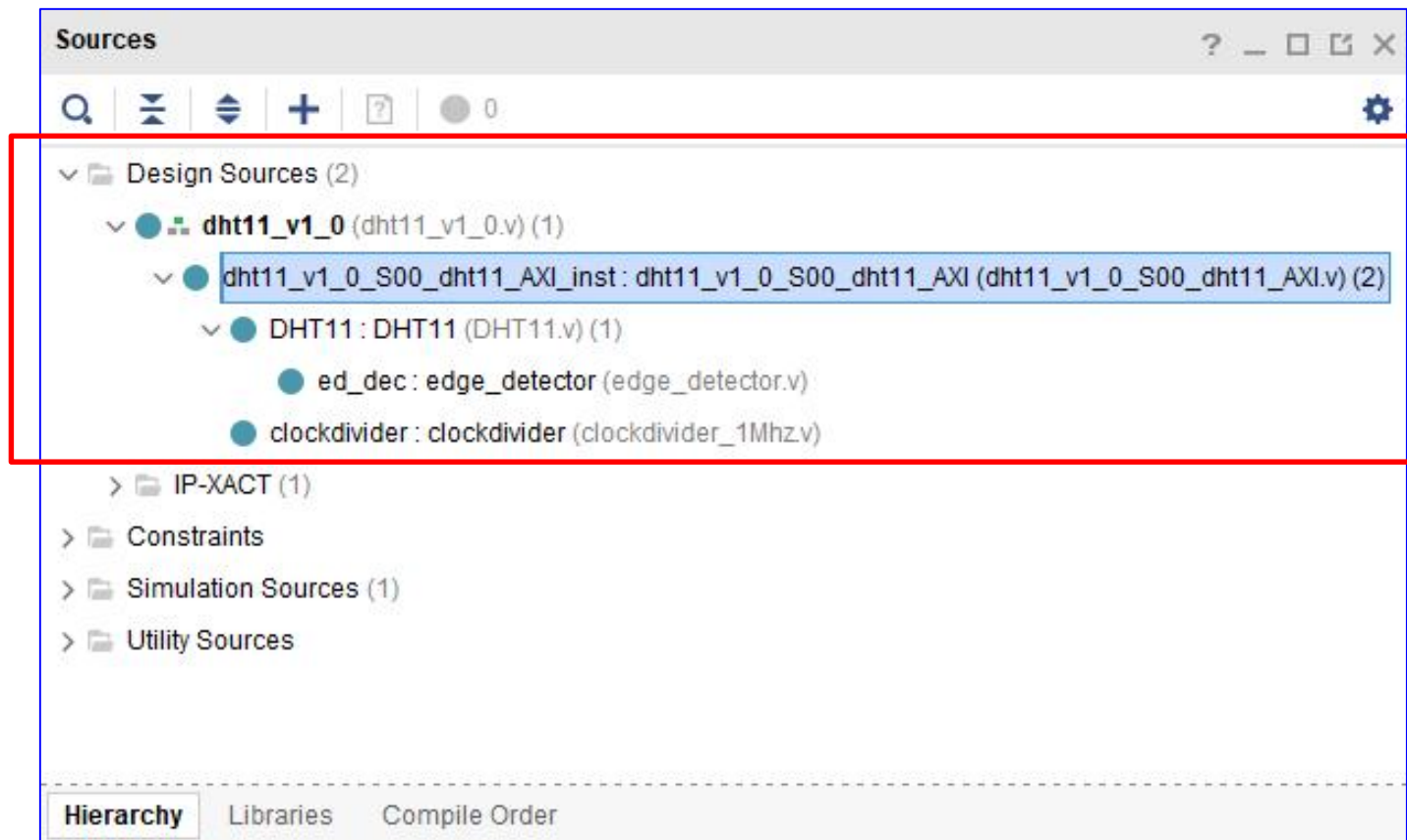
```

Package IP - dht11  x dht11_v1_0.v  x dht11_v1_0_S00_dht11_AXI.v*  x
h:/Xilinx/2023/Xilinx/ip_repo/dht11_1_0/hdl/dht11_v1_0_S00_dht11_AXI.v

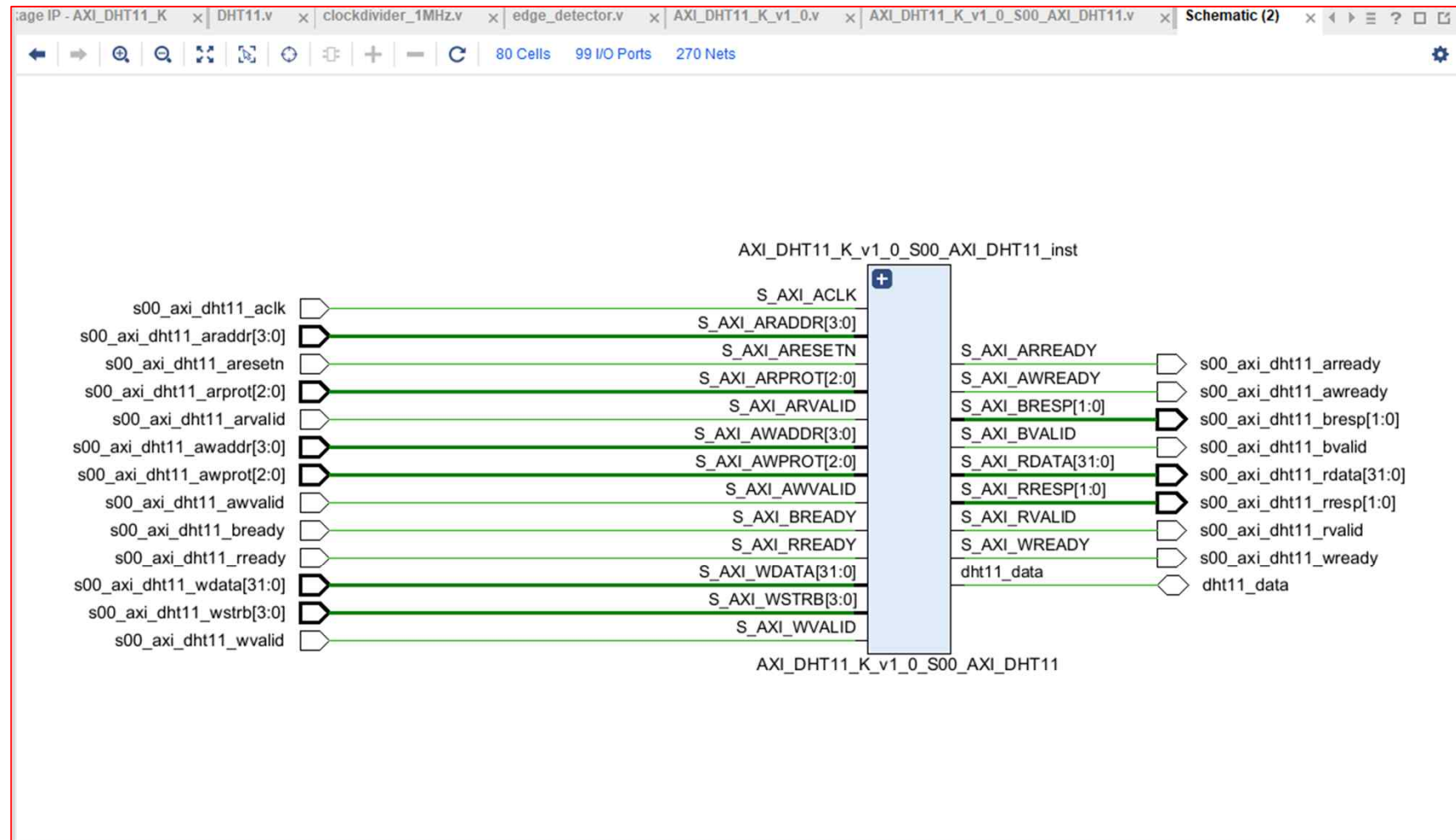
400 // Add-user logic here
401 DHT11 DHT11(
402     .clk(S_AXI_ACLK),
403     .reset(S_AXI_ARESETN),
404     .clk1Mhz(clk1Mhz),
405     .dht11_data(dht11_data),
406     .humidity10(humidity10),
407     .humidity0(humidity0),
408     .temperature10(temperature10),
409     .temperature0(temperature0)
410 );
411
412 wire [3:0] humidity10,humidity0, temperature10, temperature0;
413
414 clockdivider clockdivider(
415     .clk(S_AXI_ACLK),
416     .reset(S_AXI_ARESETN),
417     .clk1Mhz(clk1Mhz)
418 );
419 // User logic ends
420
421 endmodule
    
```

추가

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template ➔ **RTL**



➤ AXI Connect ➔ AXI Template

The screenshot displays the Xilinx Vivado IDE interface. The 'Tools' menu is open, with 'Create and Package New IP...' highlighted. The Flow Navigator on the left shows the project structure, including 'PROJECT MANAGER', 'IP INTEGRATOR', 'SIMULATION', 'RTL ANALYSIS', 'SYNTHESIS', 'IMPLEMENTATION', and 'PROGRAM AND DEBUG'. The central Hierarchy view shows the project components, including 'dht11_v1_0_S00_dht11_AXI.v'. The Source File Properties panel shows the file 'dht11_v1_0_S00_dht11_AXI.v' with the 'Enabled' checkbox checked. The Tcl Console at the bottom shows the execution of the 'create_project' and 'edit_ip_in_project' commands. The Project Summary panel on the right provides an overview of the project settings, including the project name, location, product family, and synthesis status.

Tools Menu:

- Create and Package New IP...
- Create Interface Definition...
- Enable Dynamic Function eXchange...
- Run Tcl Script...
- Property Editor
- Associate ELF Files...
- Generate Memory Configuration File...
- Compile Simulation Libraries...
- Vivado Store...
- Custom Commands
- Launch Vitis IDE
- Language Templates
- Settings...

Project Summary:

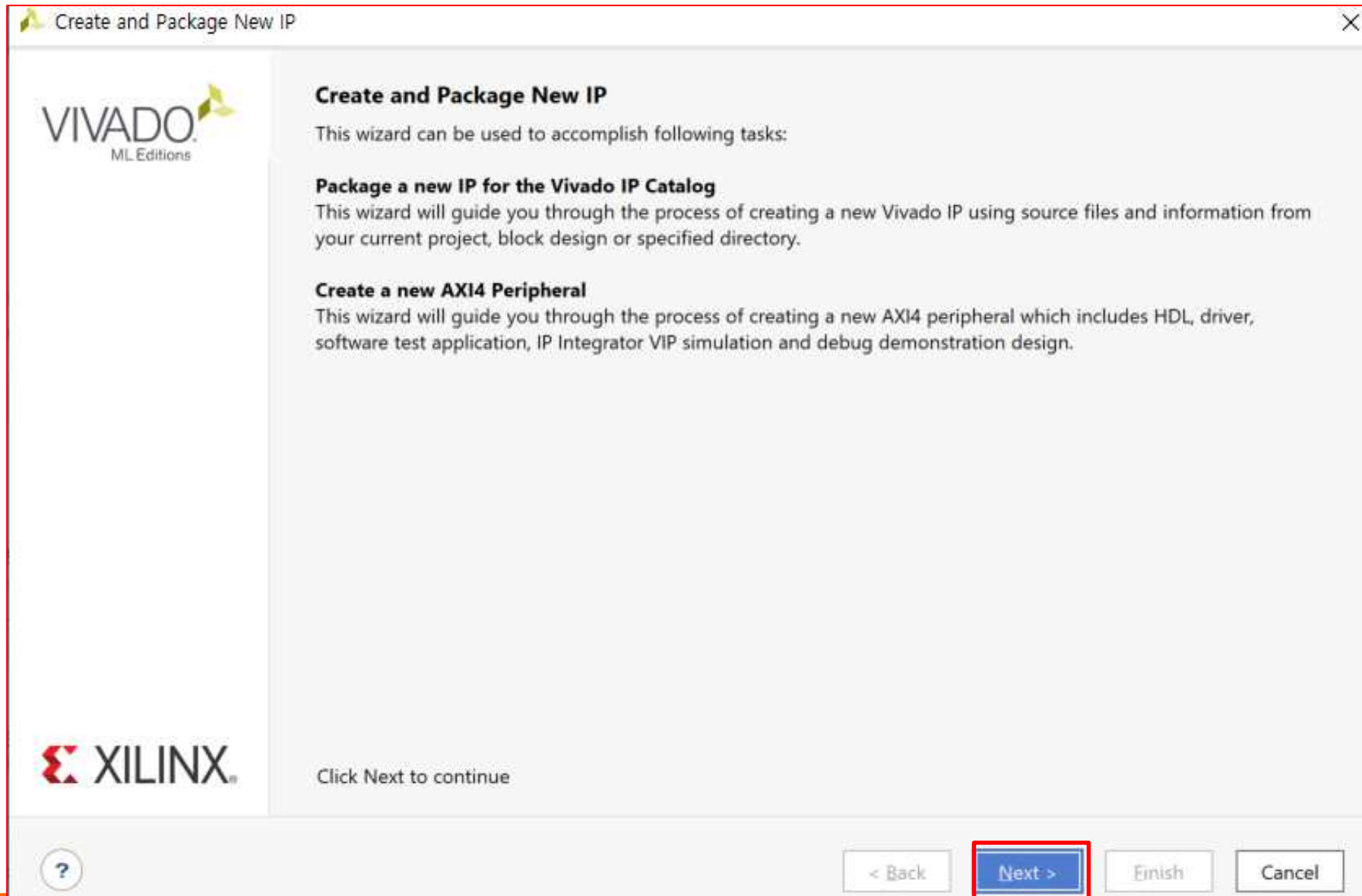
Overview	Dashboard
Settings Edit	
Project name:	dht11_v1_0_project
Project location:	h:\xilinx\2023\231221_dht11_microblaze\231221_dht11_microblaze\23
Product family:	Artix-7
Project part:	xc7a35tcpg236-1
Top module name:	dht11_v1_0
Target language:	Verilog
Simulator language:	Mixed
Synthesis	Implementation
Status:	Not started
Messages:	No errors or warnings
Part:	xc7a35tcpg236-1
Strategy:	Vivado Synthesis Defaults
Report Strategy:	Vivado Synthesis Default Reports
Incremental synthesis:	Automatically selected checkpoint

Tcl Console:

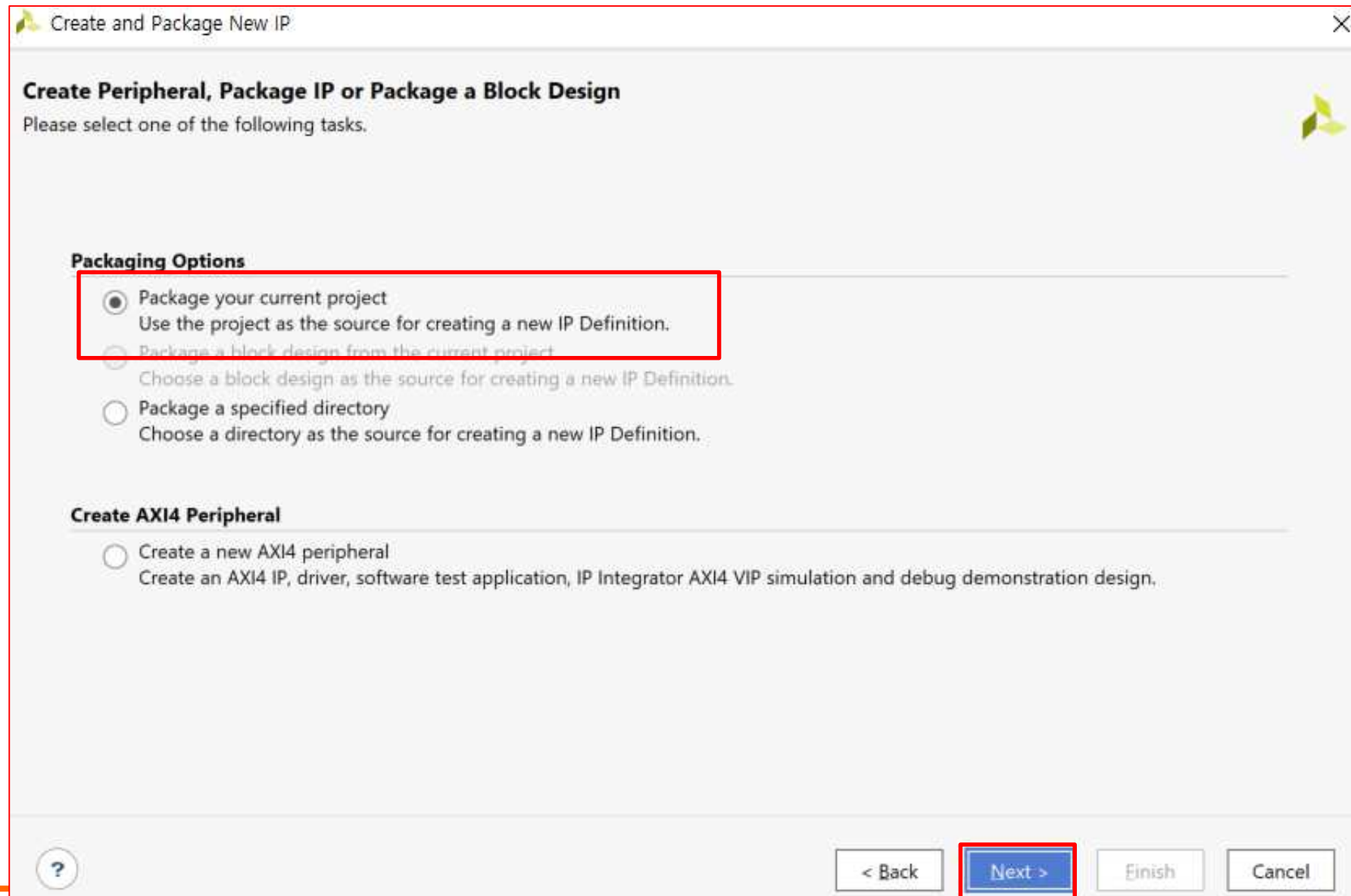
```

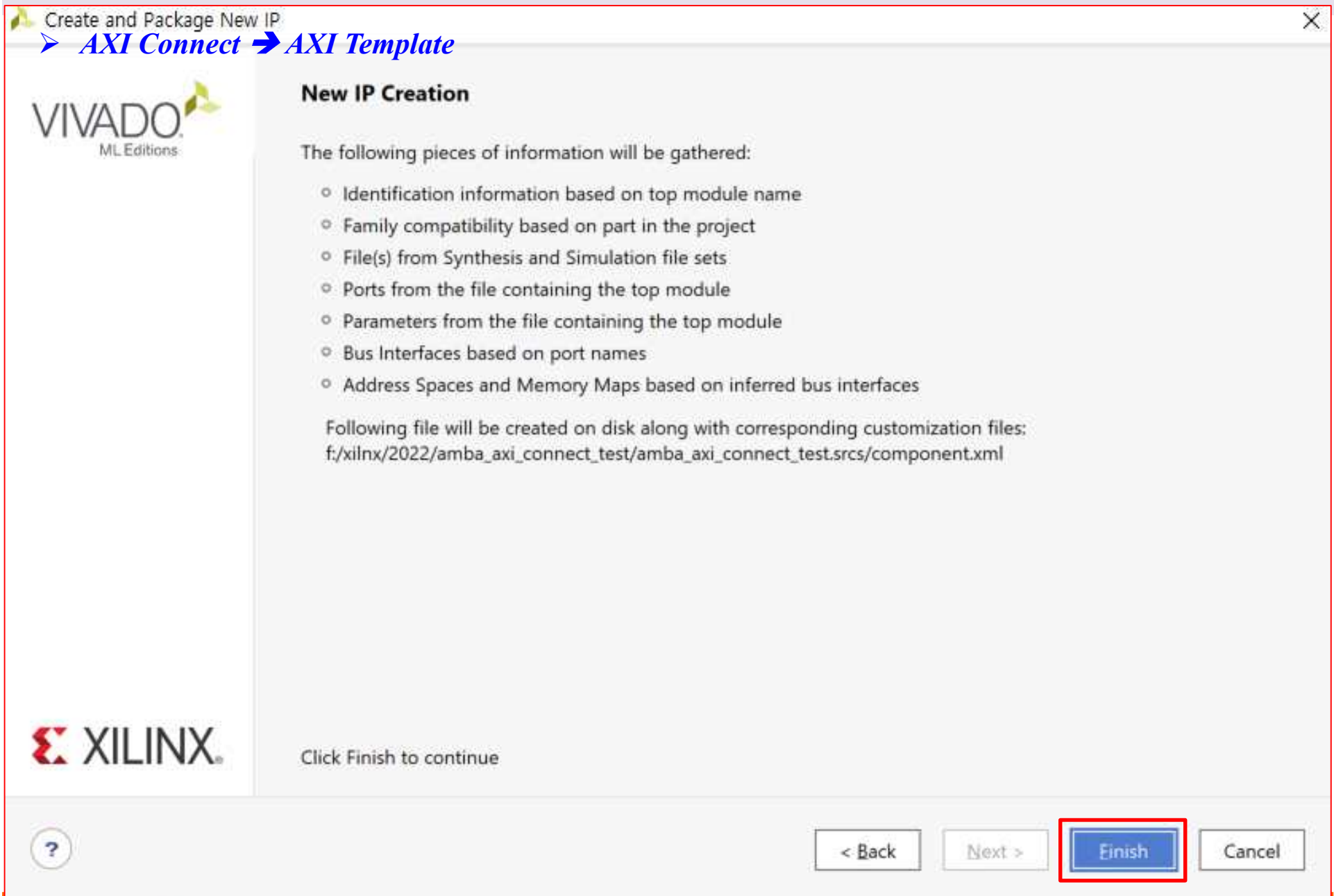
create_project: Time (s): cpu = 00:00:15 ; elapsed = 00:00:14 ; Memory (MB): peak = 4516.848 ; gain = 0.000
INFO: [IP_Flow 19-234] Refreshing IP repositories
INFO: [IP_Flow 19-1700] Loaded user IP repository 'h:\Xilinx\2023\Xilinx\ip_repo'.
INFO: [IP_Flow 19-795] Syncing license key meta-data
ipx::edit_ip_in_project: Time (s): cpu = 00:00:28 ; elapsed = 00:00:21 ; Memory (MB): peak = 4517.016 ; gain = 0.168
update_compile_order -fileset sources_1
    
```

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template





➤ AXI Connect ➔ AXI Template

MyIP 생성

The screenshot displays the Xilinx Project Manager interface for a project named 'fnd_axi_v1_0_project'. The 'Sources' pane on the left shows the project hierarchy with 'fnd_axi_v1_0' selected. The 'Packaging Steps' pane in the center lists various steps, with 'Identification' highlighted. The 'Identification' pane on the right contains the following fields:

- Vendor: xilinx.com
- Library: user
- Name: fnd_axi
- Version: 1.0
- Display name: fnd_axi_v1.0
- Description: My new AXI IP
- Vendor display name: (empty)
- Company url: (empty)
- Root directory: c:/Xilinx/ip_repo/fnd_axi_1_0
- Xml file name: c:/Xilinx/ip_repo/fnd_axi_1_0/component.xml

The 'Categories' section at the bottom shows a list with 'AXI_Peripheral'.

AMBA and AXI Connect Example

Temperature-Humidity Sensor Module

[*Create Block Design*]

AXI Connect → AXI Template

File Edit Flow Tools Reports Window Layout View Help Quick Access

Flow Navigator PROJECT MANAGER - project_1

Settings 클릭

Project Summary

Overview | Dashboard

Settings Edit

Project name: project_1
 Project location: C:/Users/parkj/test231013/project_1
 Product family: Artix-7
 Project part: Basys3 (xc7a35tcpg236-1)
 Top module name: Not defined
 Target language: Verilog
 Simulator language: Mixed

Board Part

Display name: Basys3
 Board part name: digilentinc.com:basys3:part0:1.2
 Board revision: C.0
 Connectors: No connections
 Repository path: E:/Xilinx/Vivado/2022.2/data/boards/board_files
 URL: <https://digilent.com/reference/programmable-logic/basys-3/start>
 Board overview: Basys3

Synthesis **Implementation**

Status: Not started
 Messages: No errors or warnings

Properties

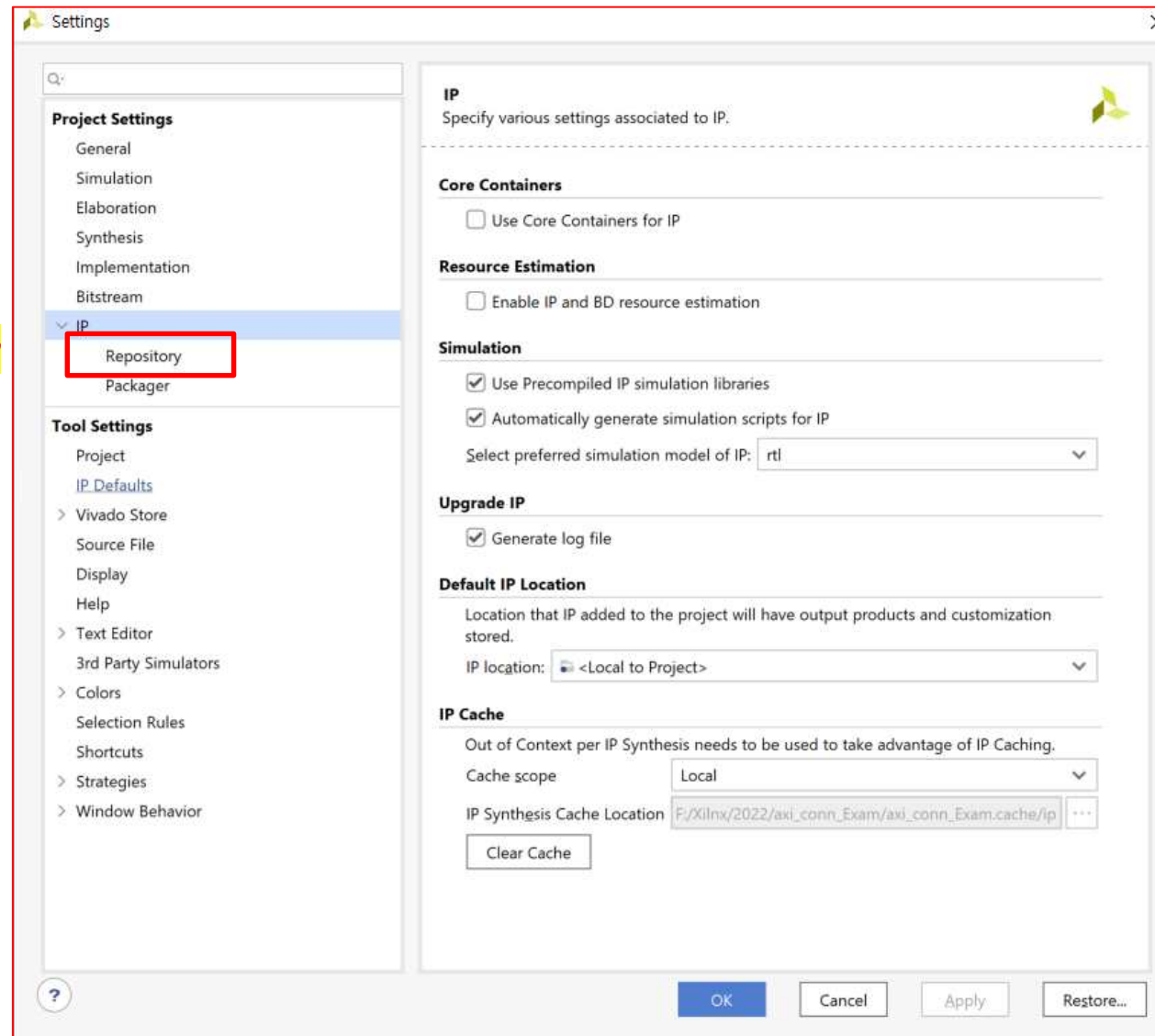
Select an object to see properties

Tcl Console Messages Log Reports **Design Runs**

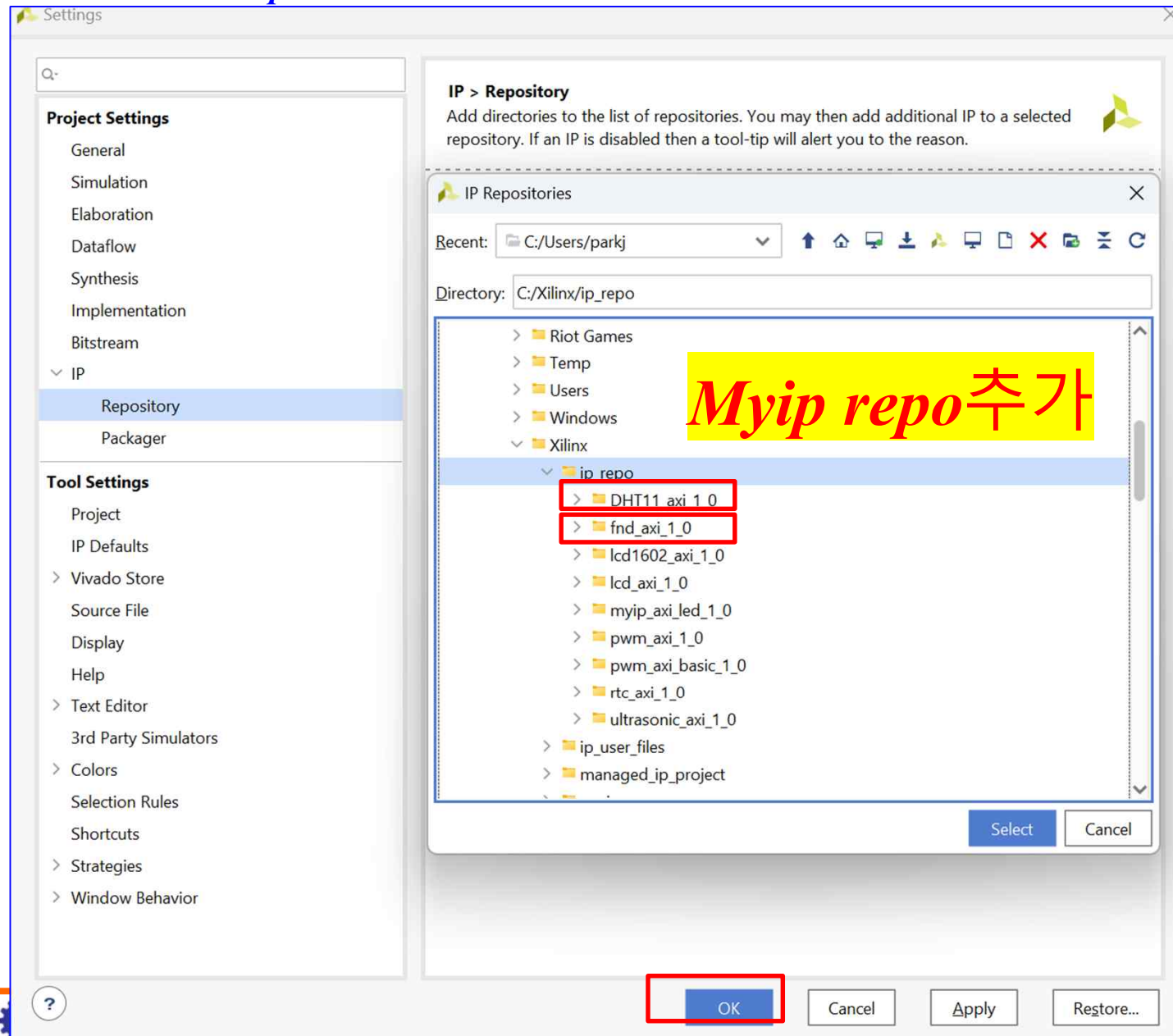
Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Not started																			Vivado Synthesis Defaults (Viva
impl_1	constrs_1	Not started																			Vivado Implementation Default

➤ AXI Connect ➔ AXI Template

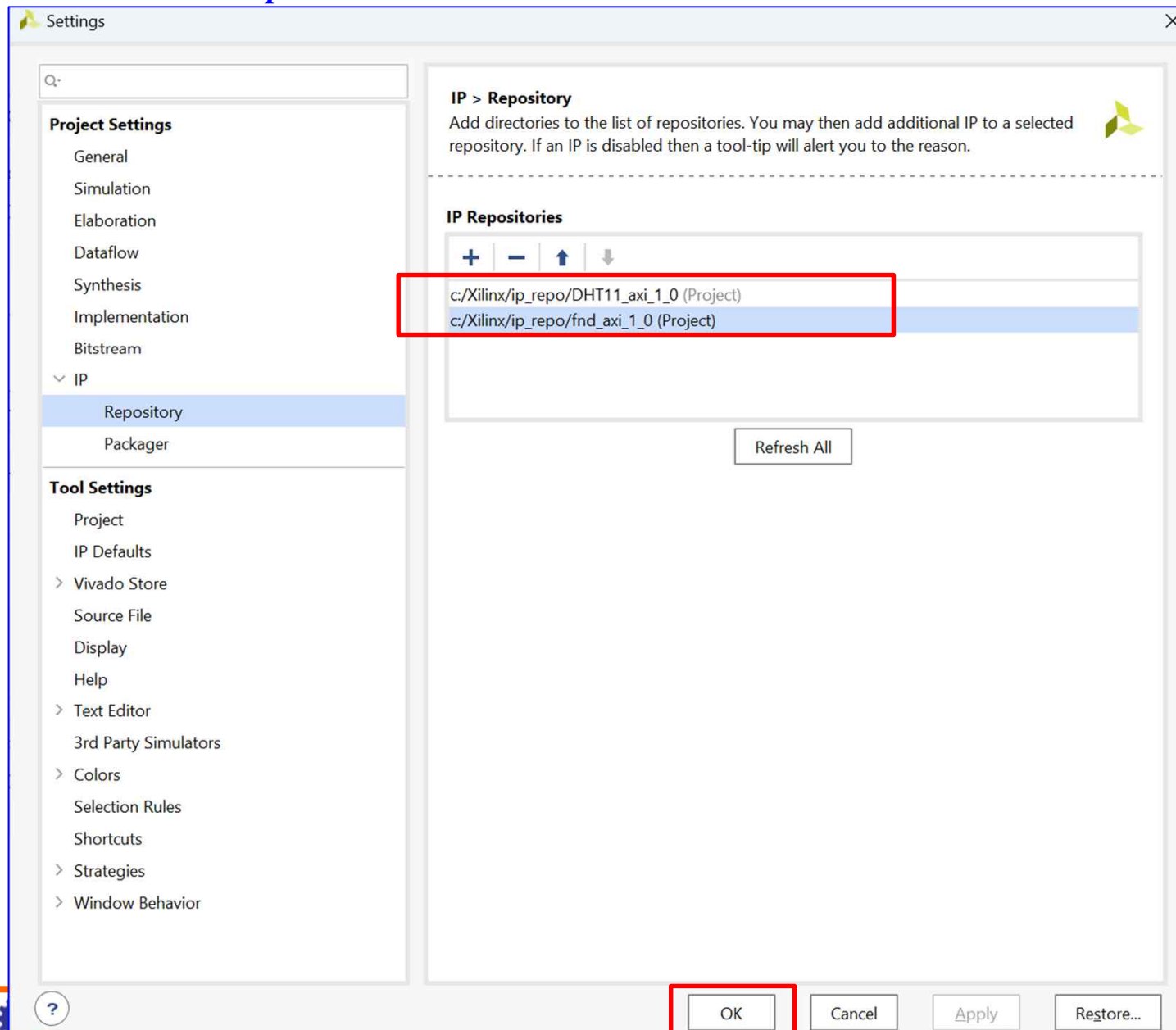
Repository



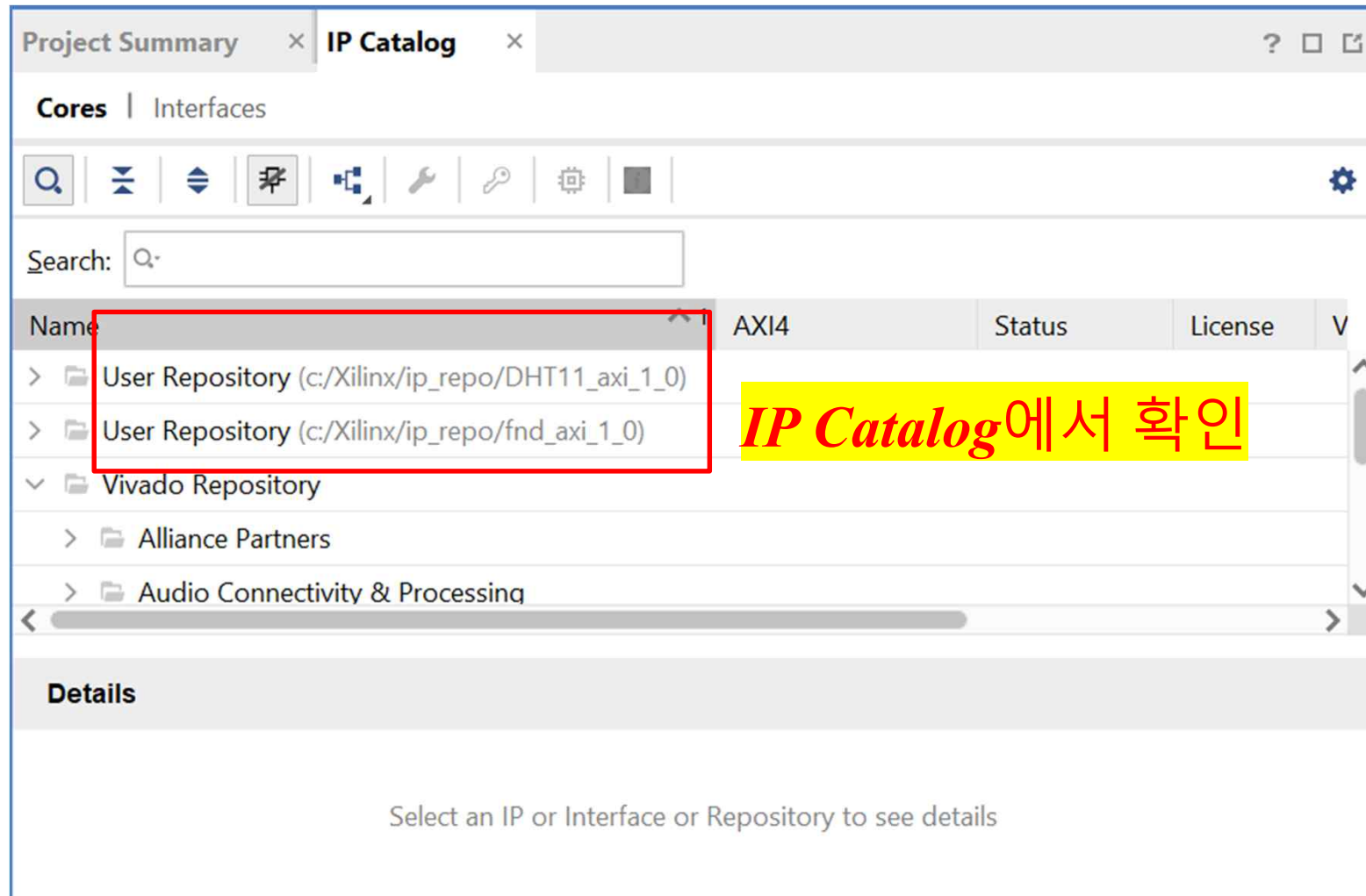
➤ AXI Connect ➔ AXI Template



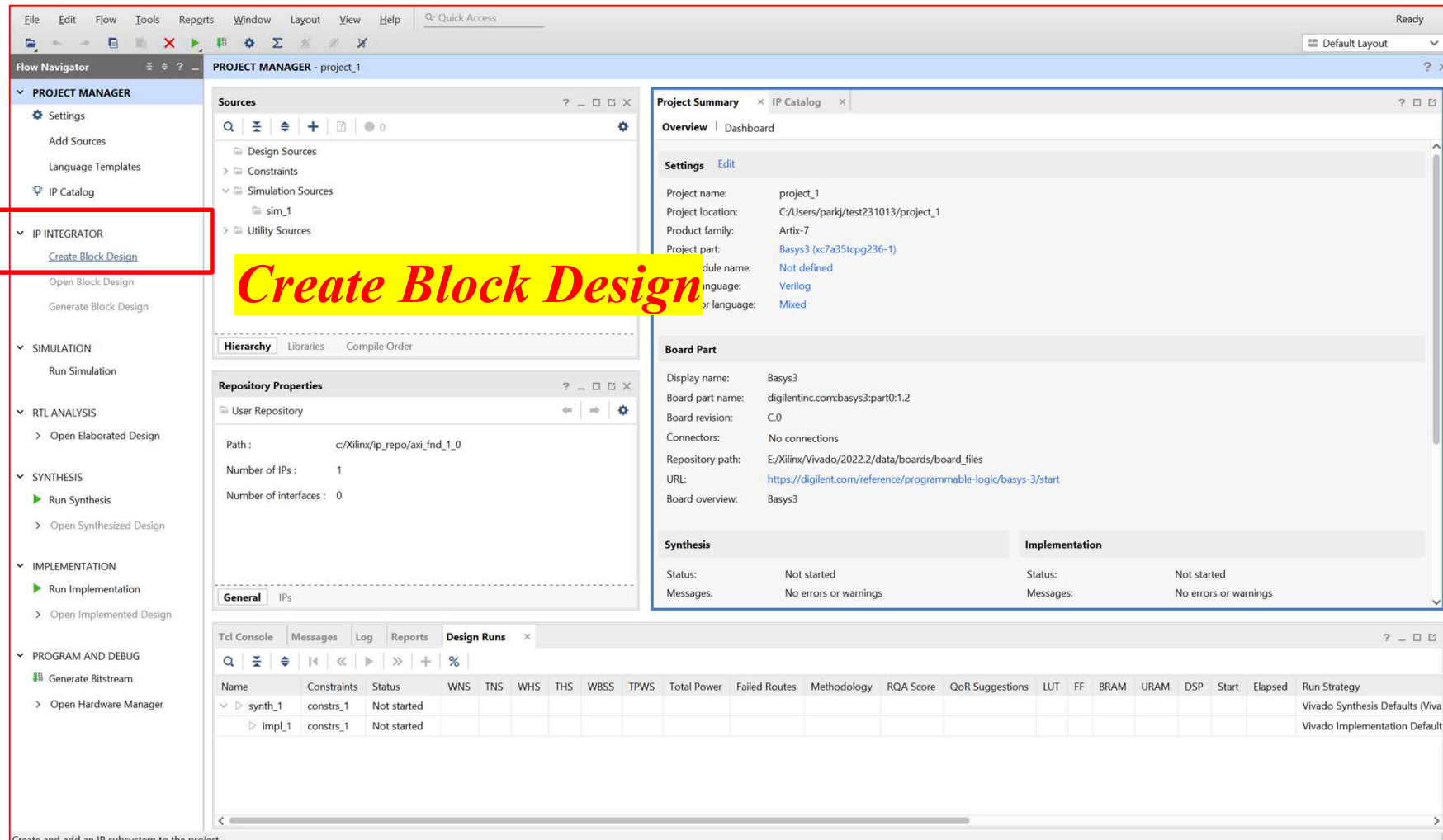
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



Create Block Design

Project Summary

Overview | Dashboard

Settings | Edit

Project name: project_1
 Project location: C:/Users/parkj/test231013/project_1
 Product family: Artix-7
 Project part: Basys3 (xc7a35tcpg236-1)
 Module name: Not defined
 Language: Verilog
 Script language: Mixed

Board Part

Display name: Basys3
 Board part name: digilentinc.com:basys3:part0:1.2
 Board revision: C.0
 Connectors: No connections
 Repository path: E:/Xilinx/Vivado/2022.2/data/boards/board_files
 URL: <https://digilent.com/reference/programmable-logic/basys-3/start>
 Board overview: Basys3

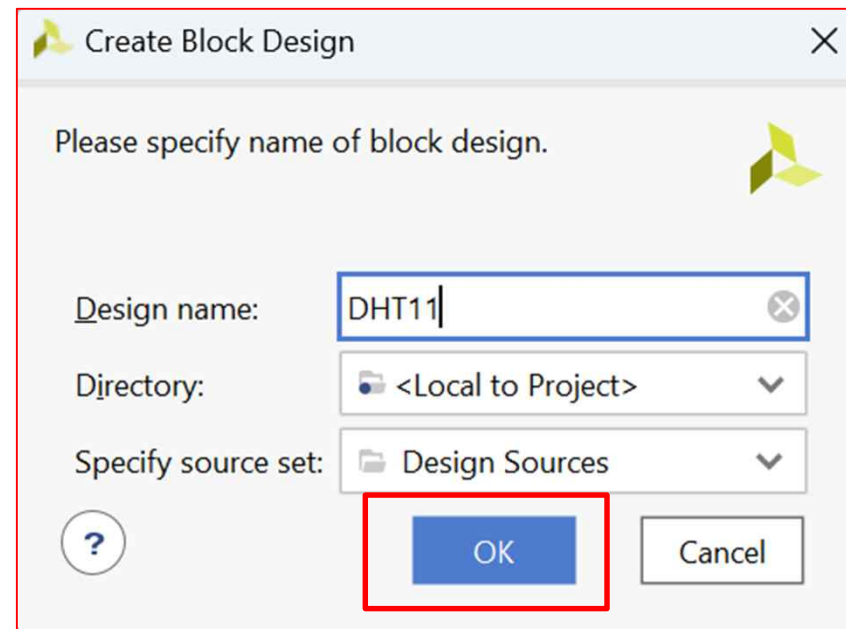
Synthesis | **Implementation**

Status: Not started
 Messages: No errors or warnings

Design Runs

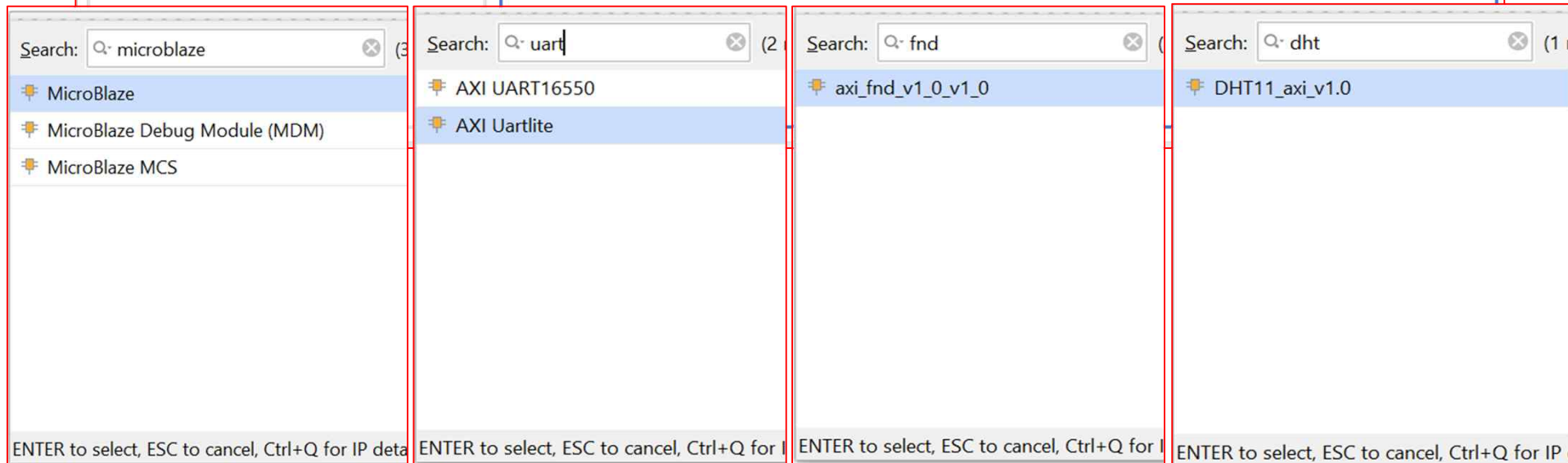
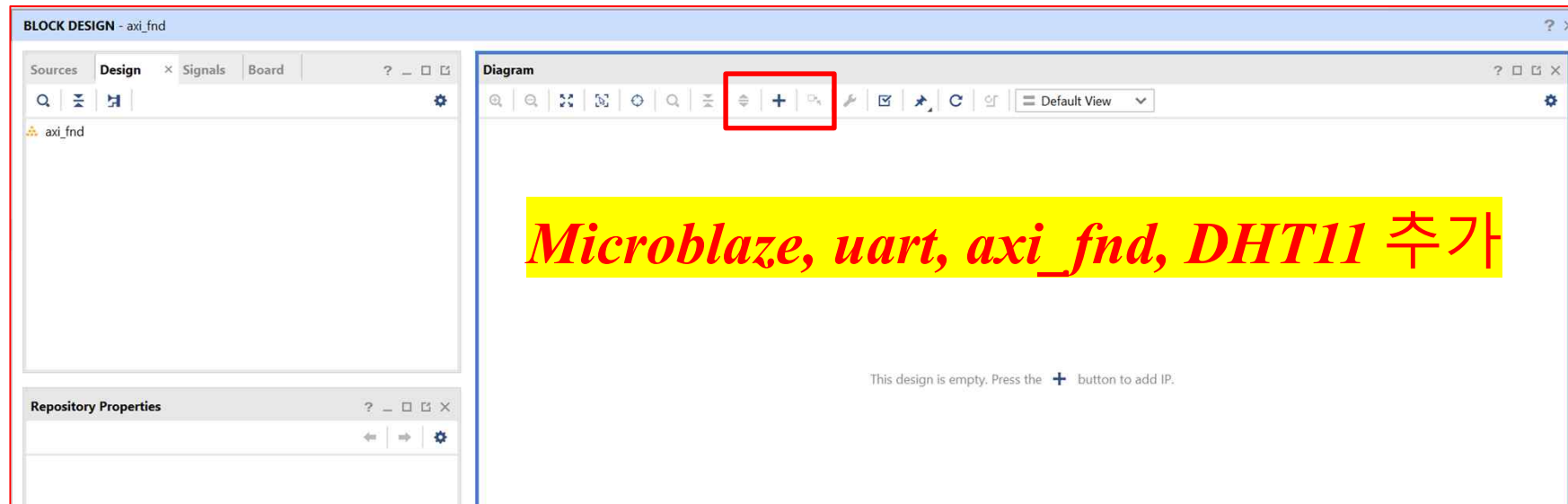
Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Not started																			Vivado Synthesis Defaults (Viva
impl_1	constrs_1	Not started																			Vivado Implementation Default

➤ AXI Connect ➔ AXI Template

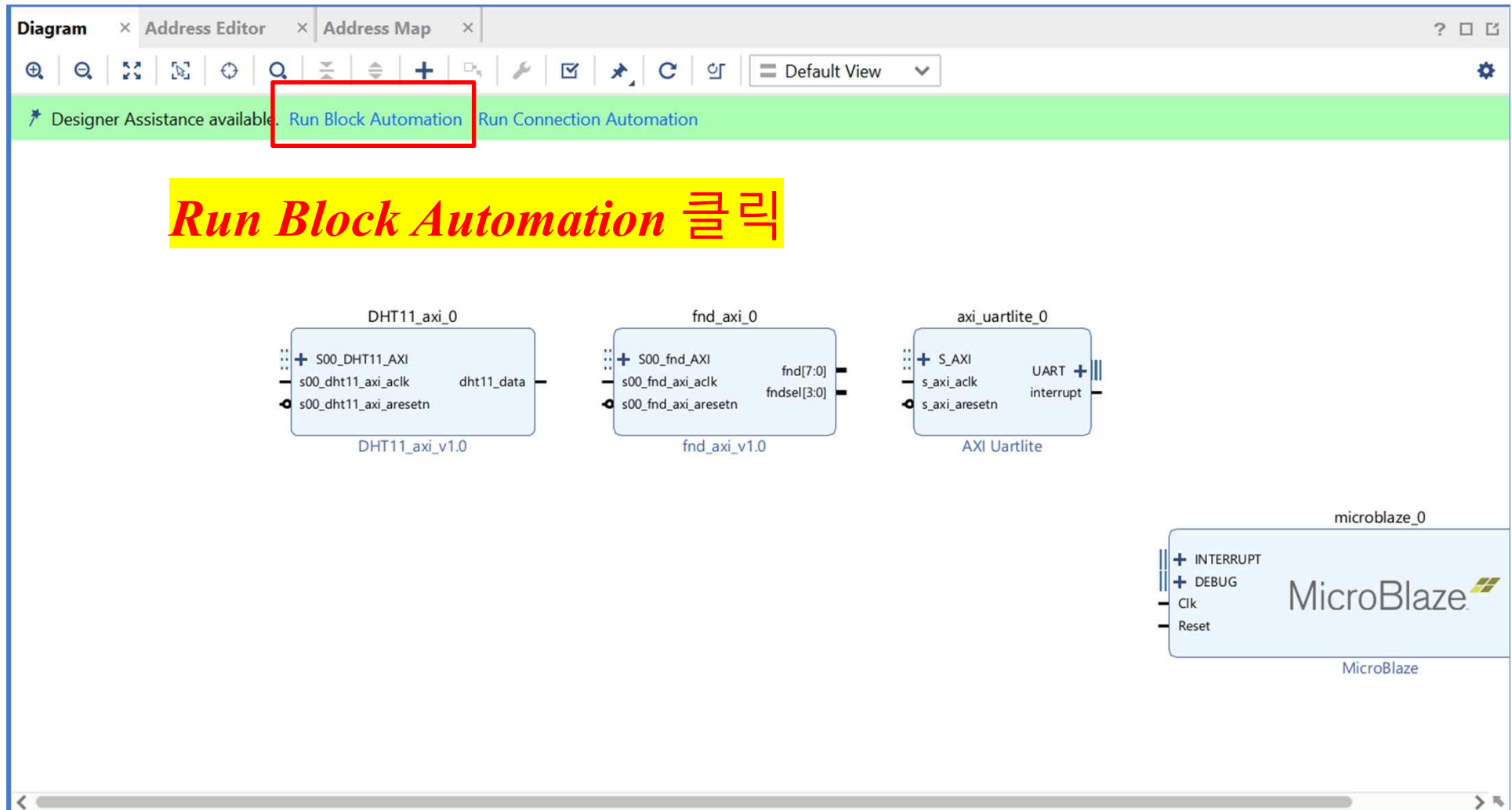


이름 설정 후 OK

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

Automatically make connections in your design by checking the boxes of the blocks to connect. Select a block on the left to display its configuration options on the right.



✓ All Automation (1 out of 1 selected)

✓ microblaze_0

Description

MicroBlaze connection automation generates local memory of selected size, and caches can be configured. MicroBlaze Debug Module, Peripheral AXI Interconnect, Interrupt Controller, a clock source, Processor System Reset are added and connected as needed. A preset MicroBlaze configuration can also be selected.

Information about the options can be found in the tooltips.

Options

Preset None ▼

Local Memory 128KB ▼

Local Memory ECC None ▼

Cache Configuration None ▼

Debug Module Debug Only ▼

Peripheral AXI Port Enabled ▼

☐ Interrupt Controller

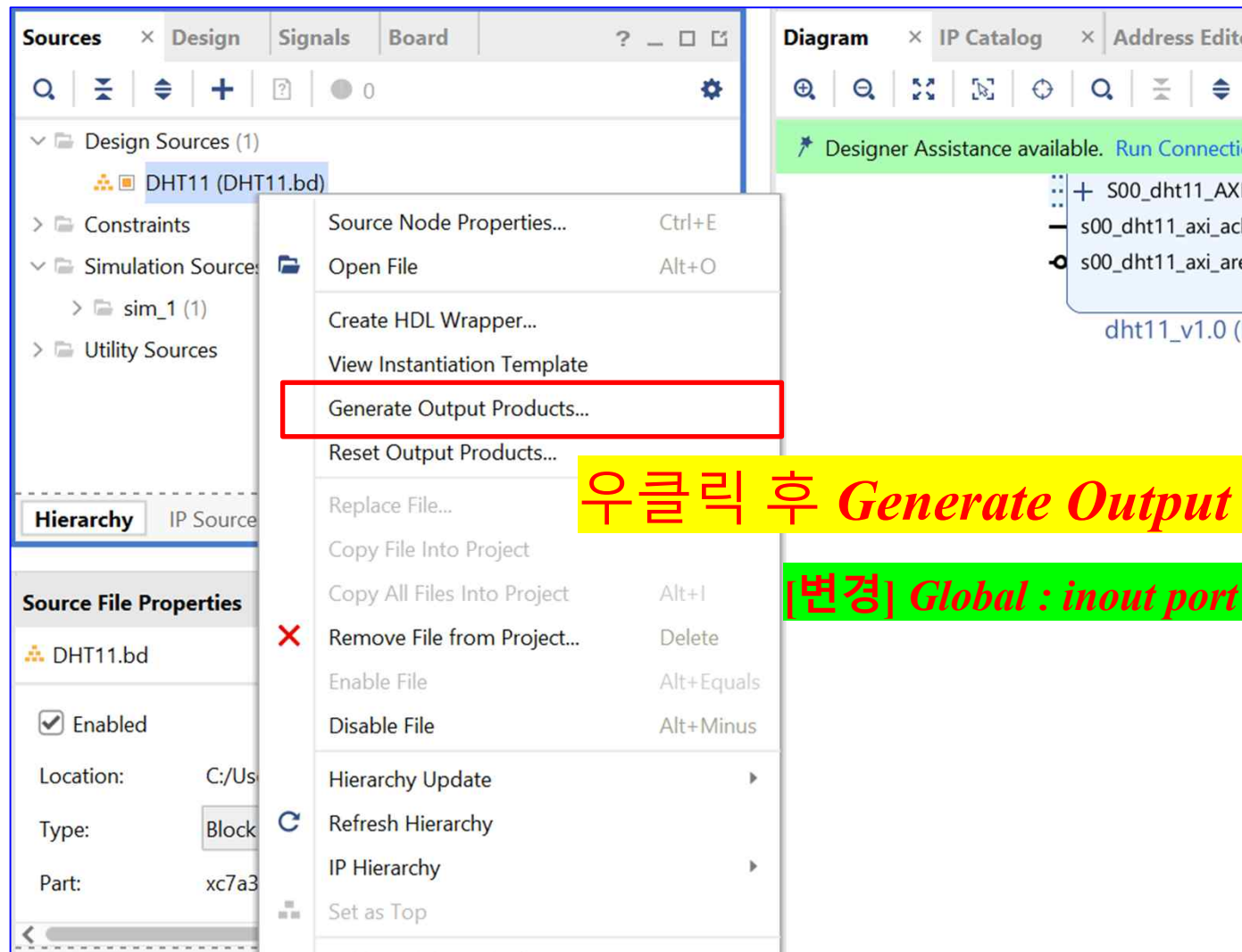
Clock Connection New Clocking Wizard ▼



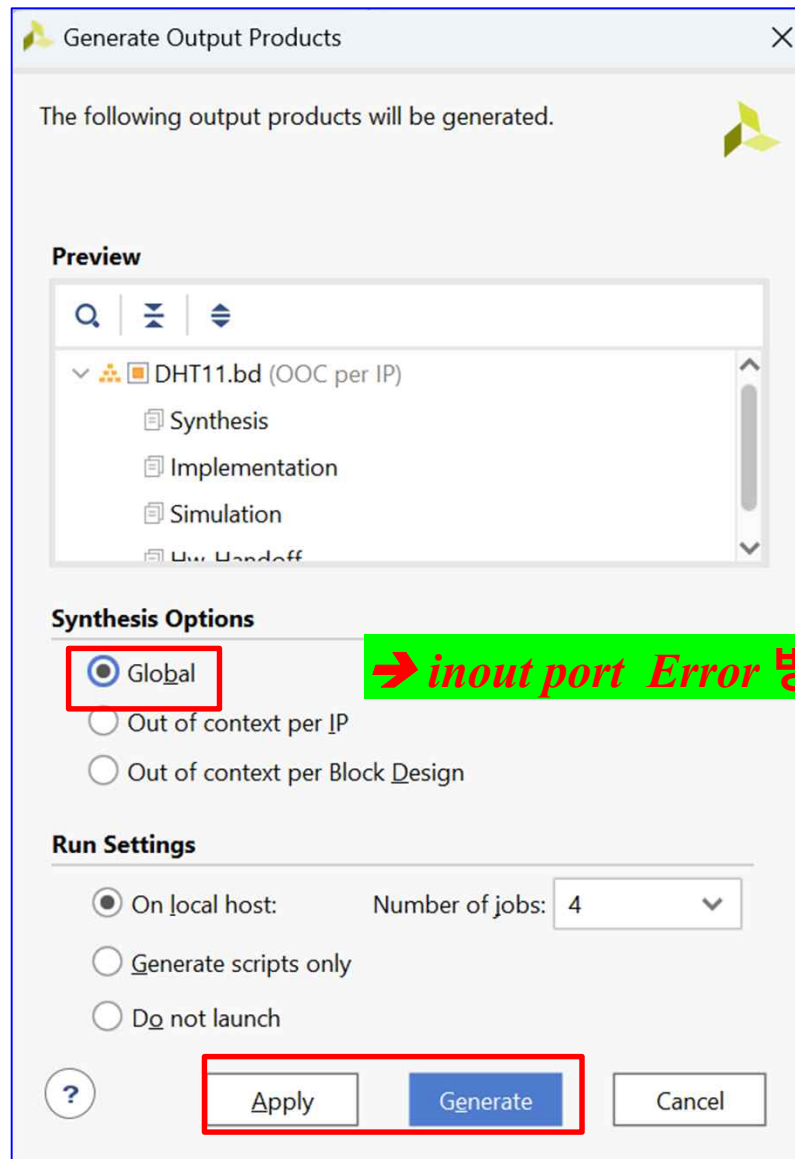
OK

Cancel

➤ AXI Connect ➔ AXI Template



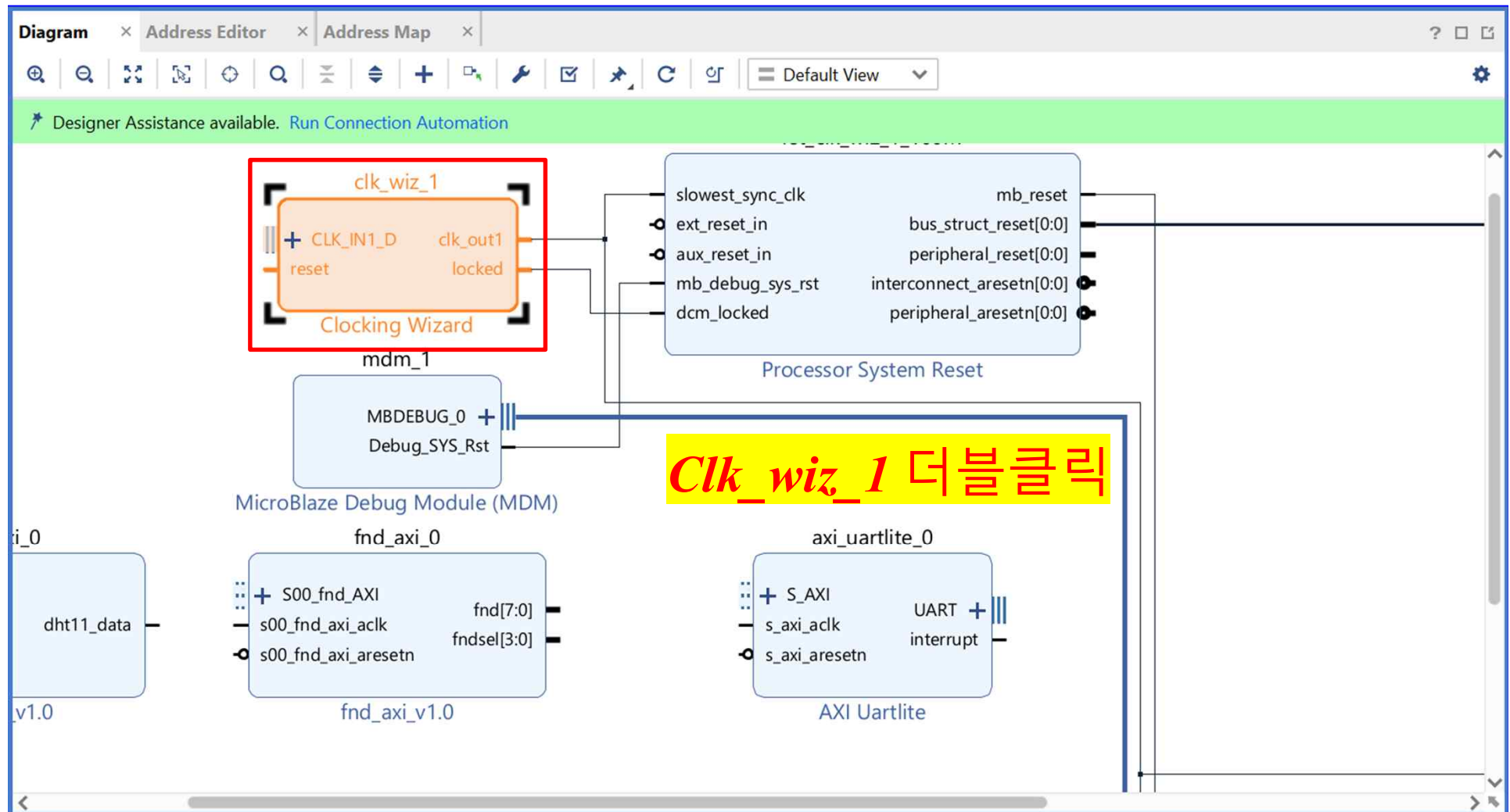
➤ AXI Connect ➔ AXI Template



➔ inout port Error 방지

Synthesis Options를 Global로
변경 후 Generate

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template Clocking Wizard (6.0)

Documentation IP Location

IP Symbol **Resource**

☒ Show disabled ports

- + s_axi_lite
- + CLK_IN1_D
- + CLK_IN2_D
- + CLKFB_IN_D
- CLKFB_OUT_D
- s_axi_adck
- s_axi_aresetn
- reset
- resetn
- ref_clk
- user_clk0
- user_clk1
- user_clk2
- user_clk3
- clk_in1
- clk_stop[3:0]
- clk_glitch[3:0]
- interrupt
- clk_oor[3:0]
- clk_out1
- locked

Component Name clk_wiz_1

Board **Clocking Options** Output Clocks MMCM Settings Summary

Clock Monitor

☐ Enable Clock Monitoring

Primitive

☒ MMCM ☐ PLL

Clocking Features

☒ Frequency Synthesis ☐ Minimize Power
☒ Phase Alignment ☐ Spread Spectrum
☐ Dynamic Reconfig ☐ Dynamic Phase Shift
☐ Safe Clock Startup

Jitter Optimization

☒ Balanced
☐ Minimize Output Jitter
☐ Maximize Input Jitter filtering

Dynamic Reconfig Interface

Options ☐ Phase Duty Cycle Config ☐ Write DRP registers
☒ AXI4Lite ☐ DRP

Input Clock Information

	Input Clock	Port Name	Input Frequency(MHz)		Jitter Options	Input Jitter	Source
☒	Primary	clk_in1	AUTO 100.000	10.000 - 800.000	UI	0.010	Single ended clock capable p
☐	Secondary	clk_in2	AUTO 100.000	60.000 - 120.000		0.010	Single ended clock capable p

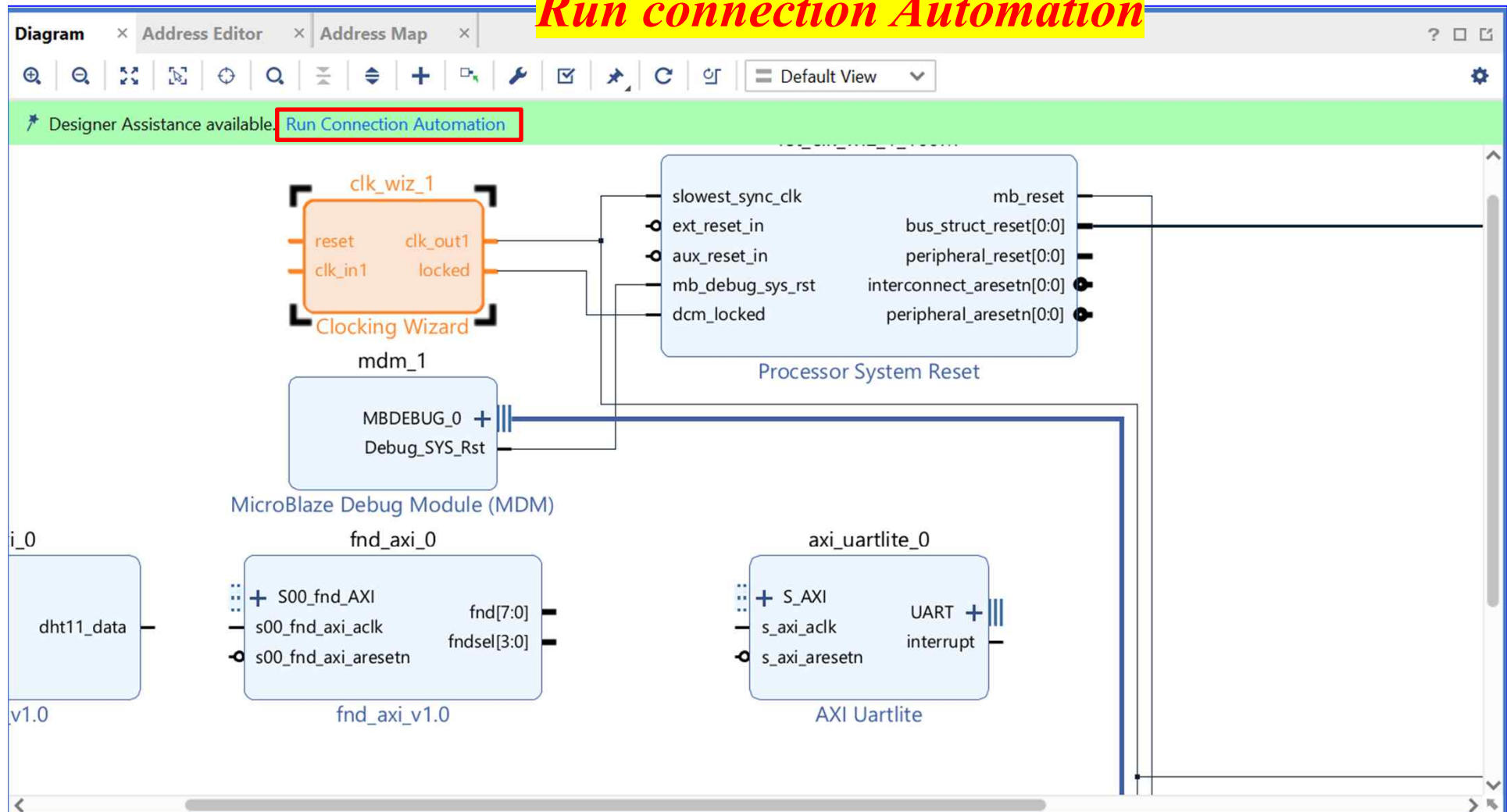
Single ended clock 변경 후 OK

OK

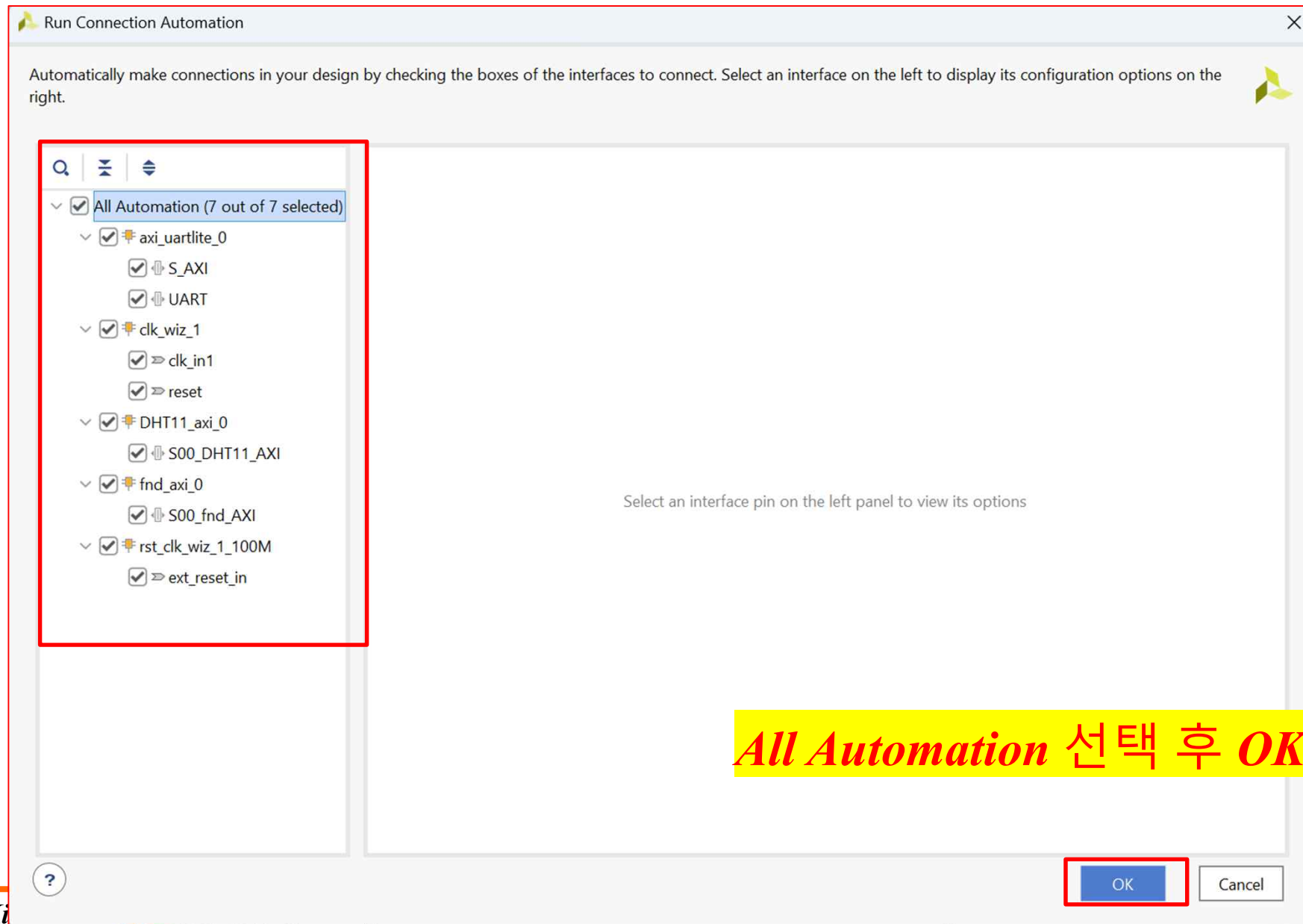
Cancel

➤ AXI Connect ➔ AXI Template

Run connection Automation

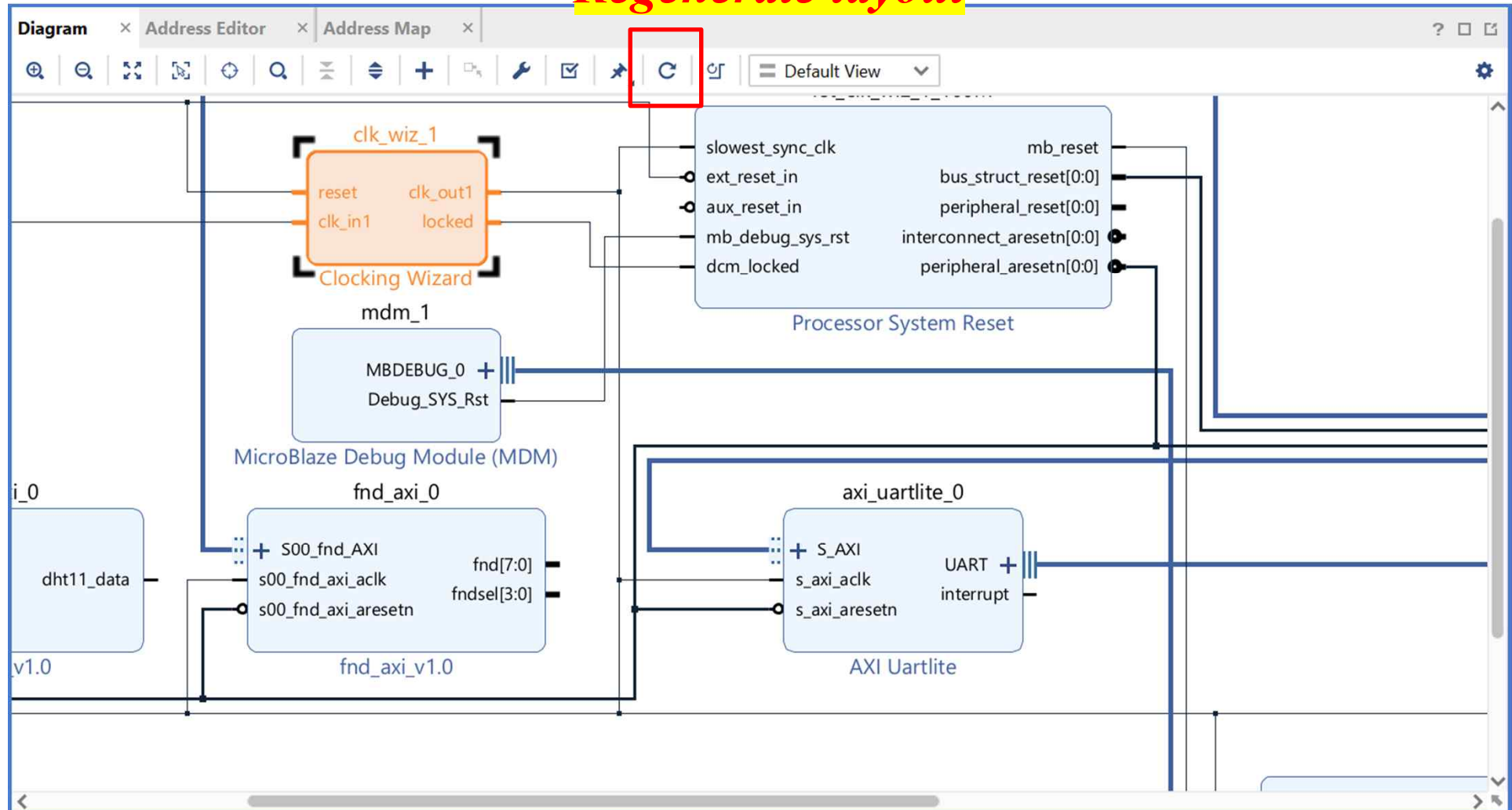


➤ AXI Connect ➔ AXI Template

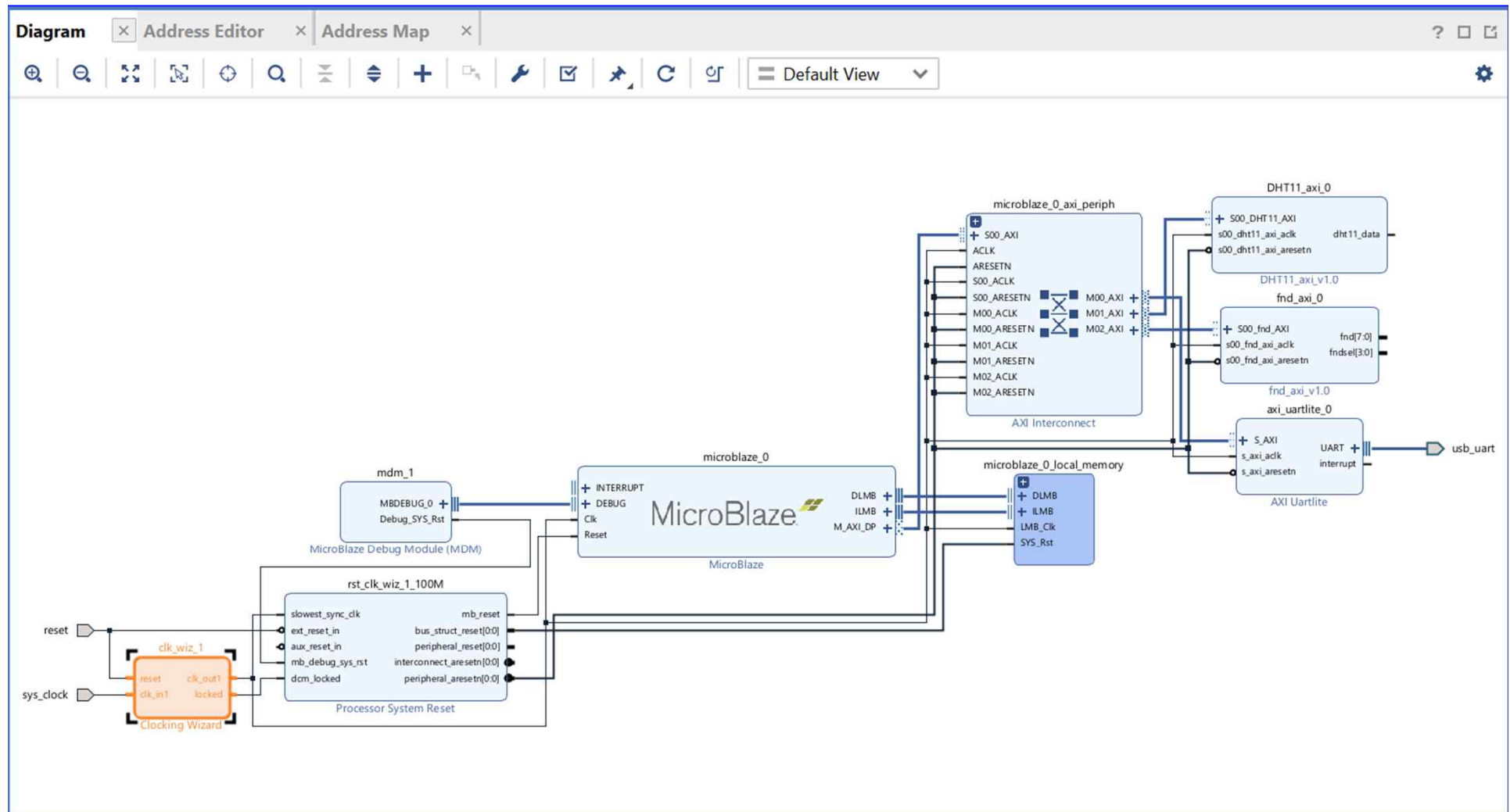


➤ AXI Connect ➔ AXI Template

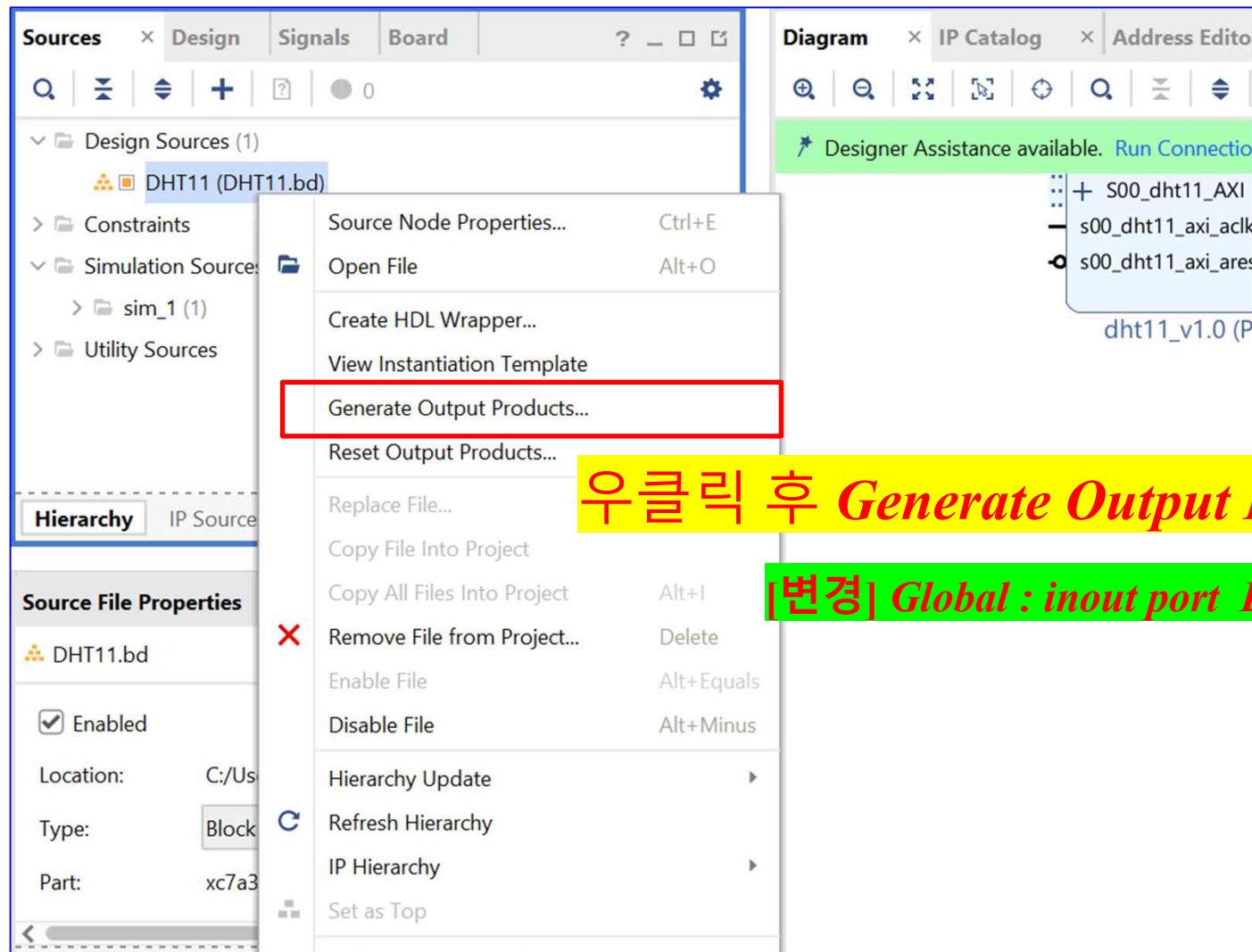
Regenerate layout



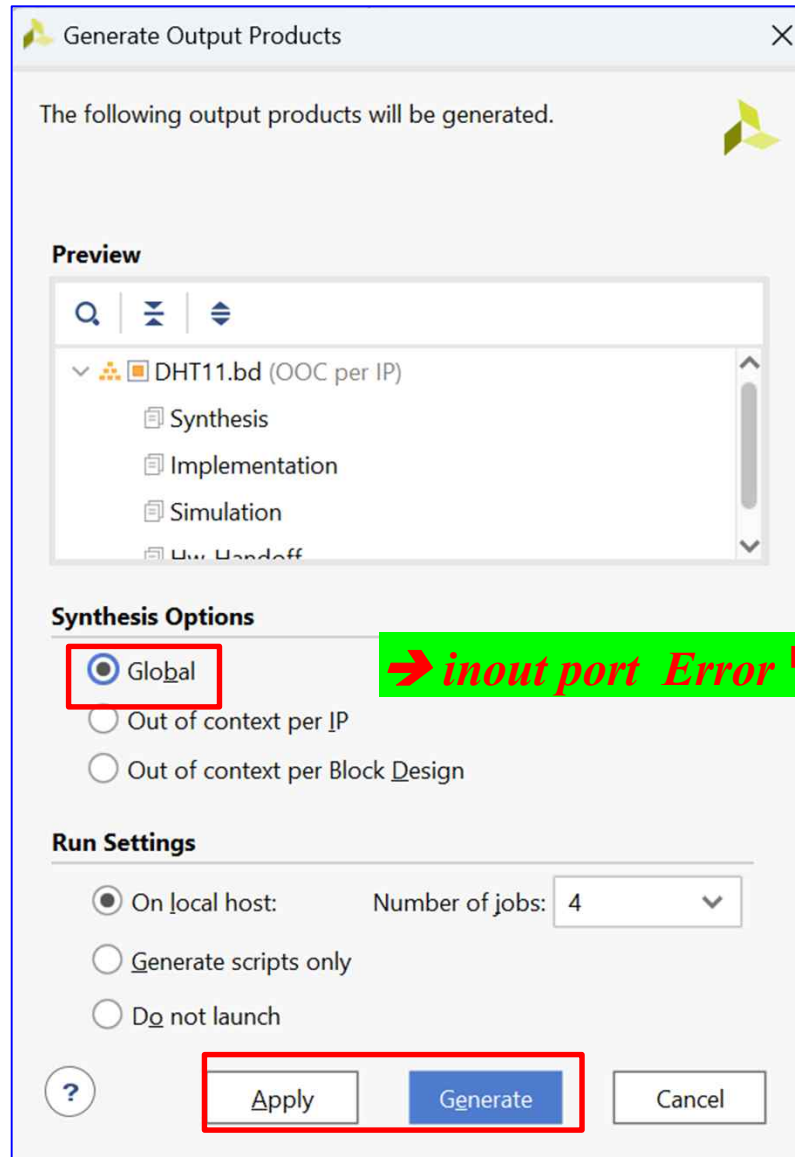
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



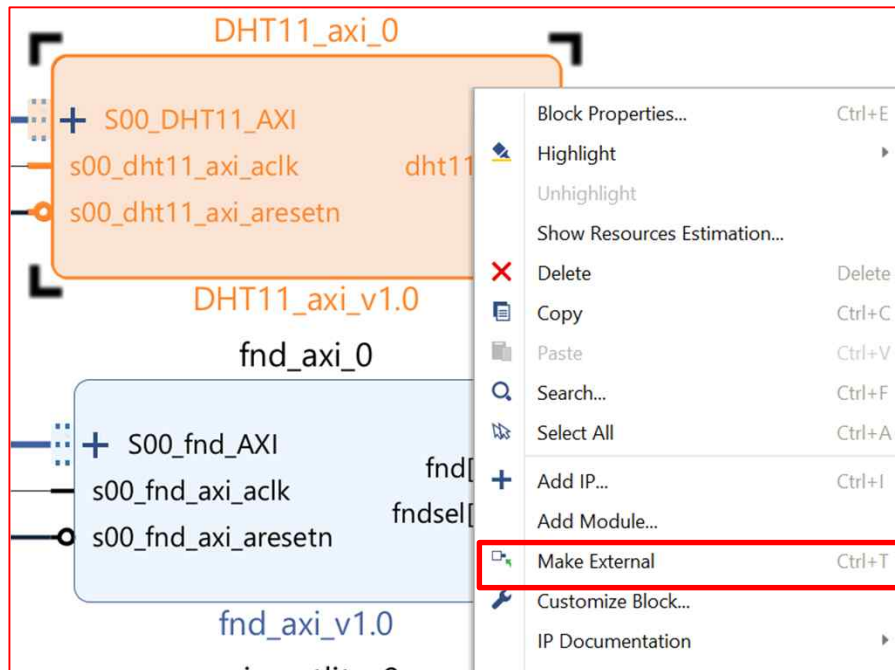
➤ AXI Connect ➔ AXI Template



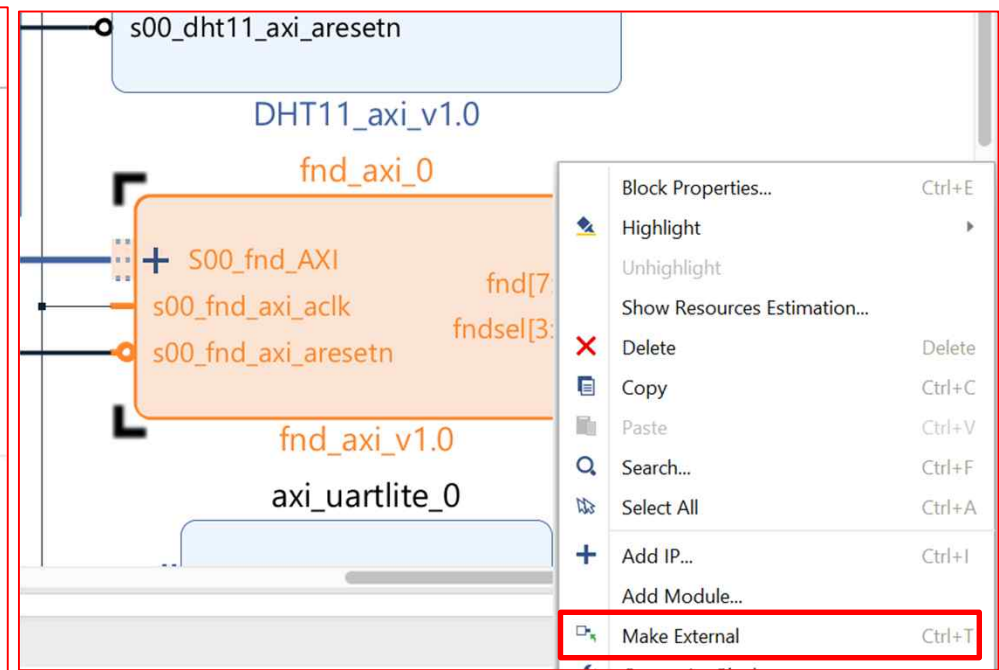
➔ inout port Error 방지

Synthesis Options를 Global로
변경 후 Generate

➤ AXI Connect ➔ AXI Template

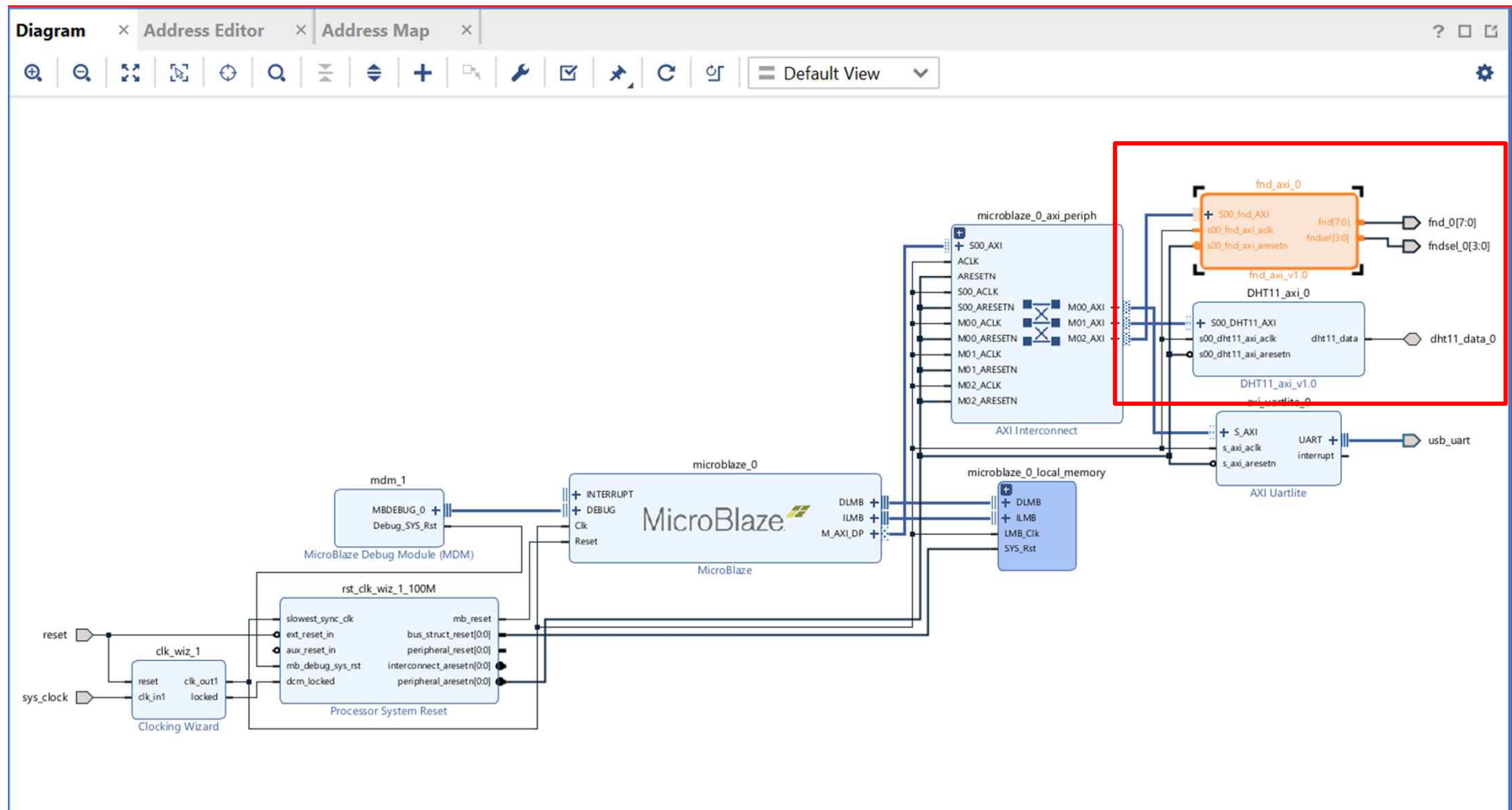


Make External

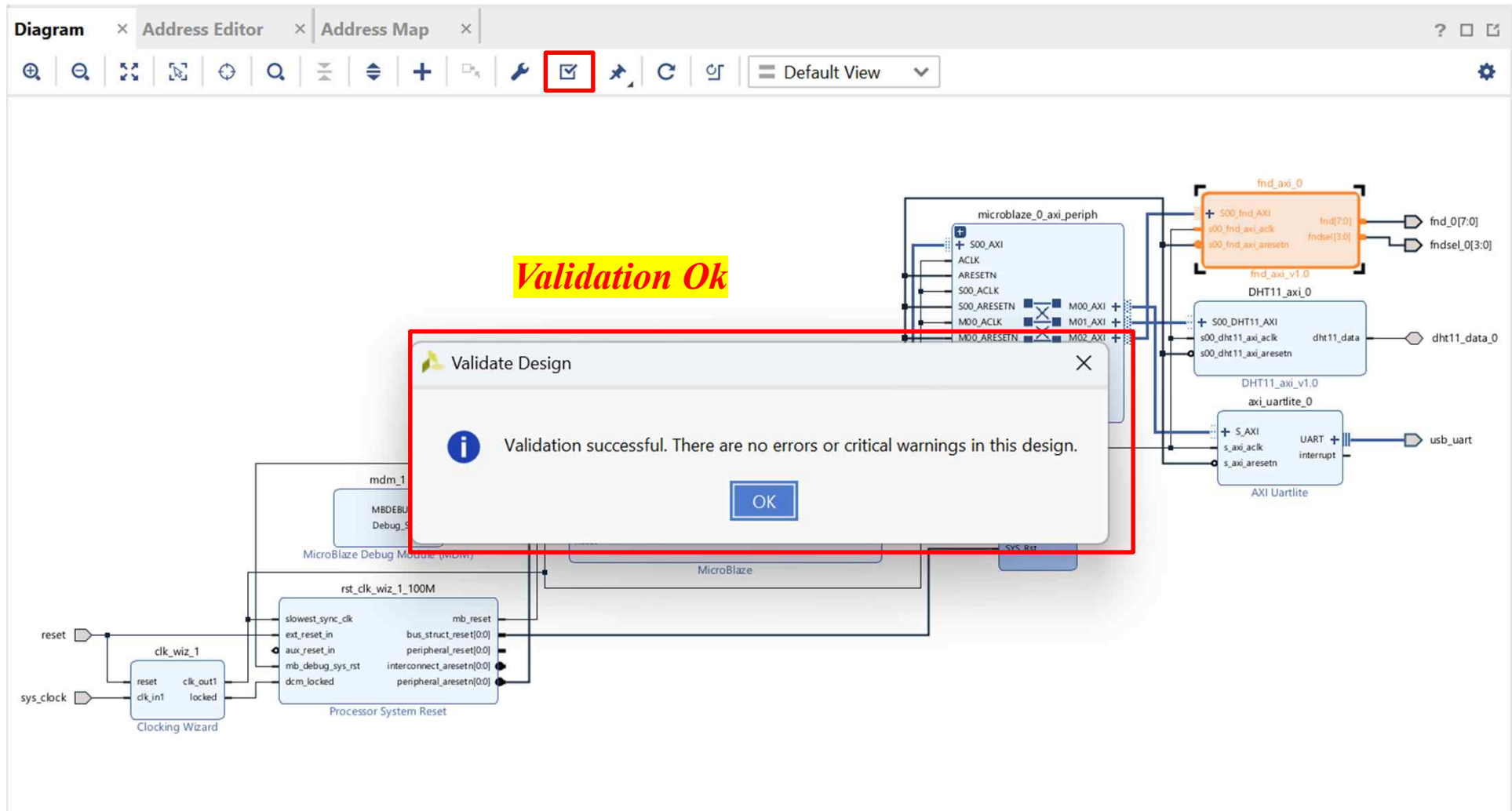


Make External

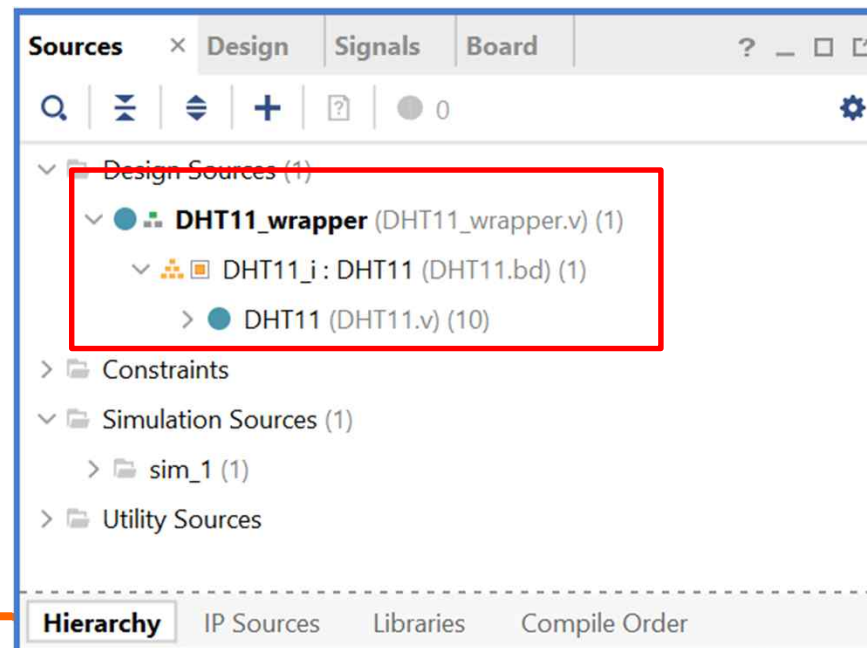
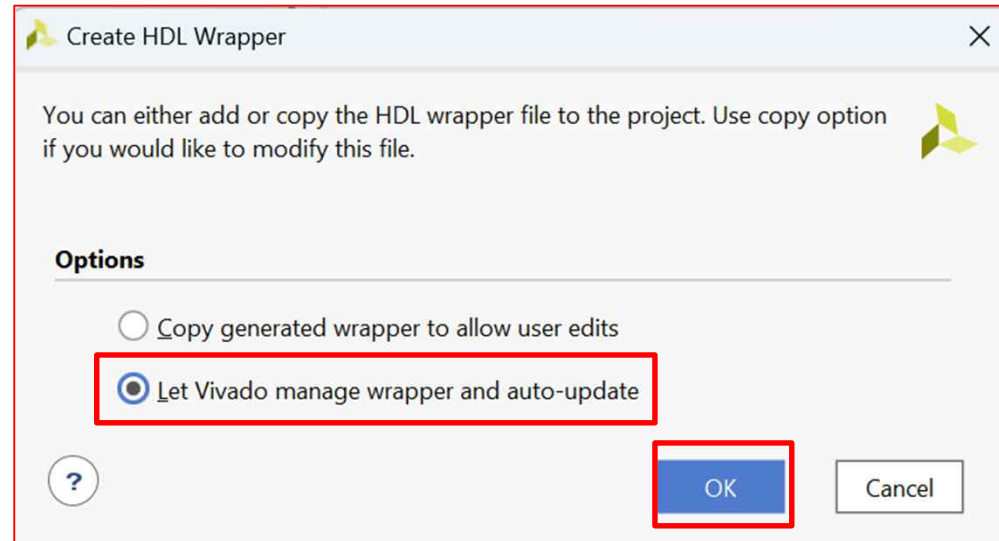
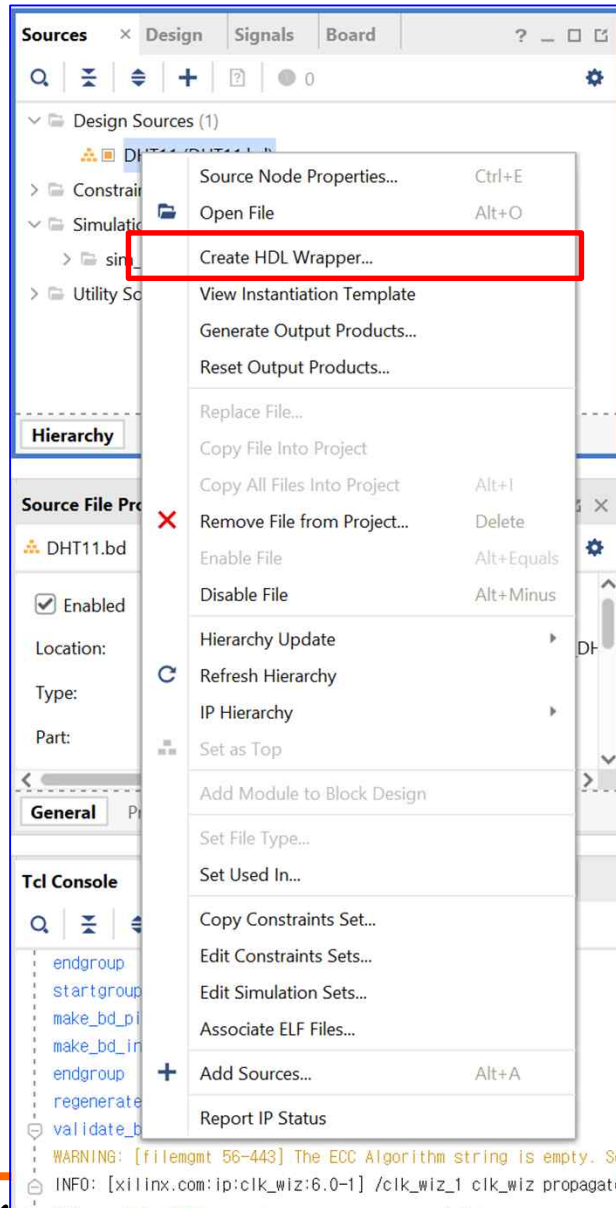
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

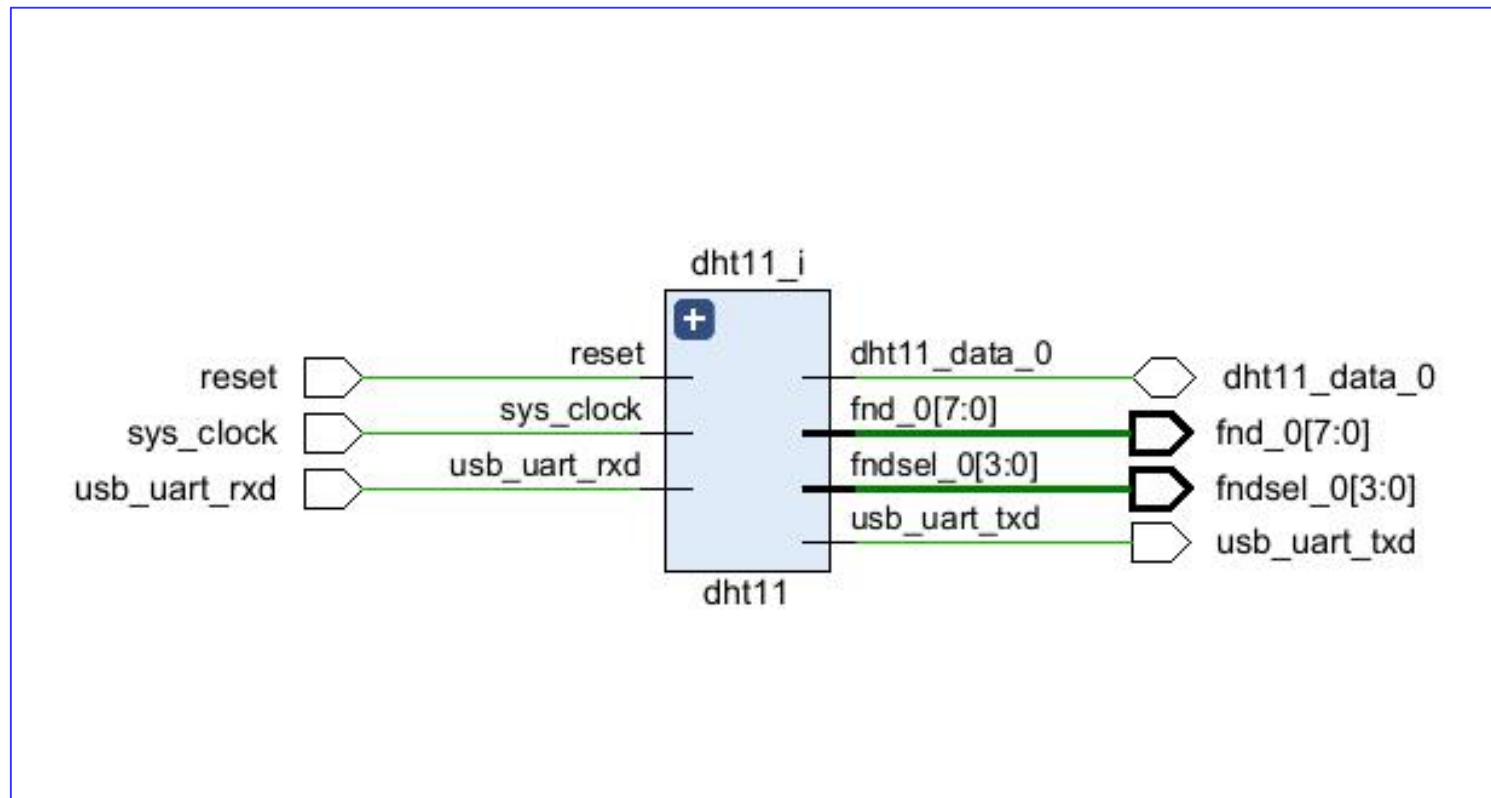


➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

RTL ➔ Schematic



AXI Connect → AXI Template

Add Sources

This guides you through the process of adding and creating sources for your project

- ☒ Add or create constraints **Xdc 파일 추가**
- ☐ Add or create design sources
- ☐ Add or create simulation sources

Source File Properties

DHT11_wrapper.v

☒ Enabled

Location: c:/Users/parkj/2

Type: Verilog

Library: xil_defaultlib

Tcl Console

WARNING: [filegmt 56-443]

WARNING: [filegmt 56-443]

XILINX

< Back Next > Finish Cancel

➤ AXI Connect ➔ AXI Template

▪ XDC Constraints

Basys-3-Master.xdc

1. `set_property -dict { PACKAGE_PIN W5 IOSTANDARD LVCMOS33 } [get_ports sys_clock]`
2. `create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports sys_clock]`
3. **## Switches**
4. `set_property -dict { PACKAGE_PIN R2 IOSTANDARD LVCMOS33 } [get_ports {reset}]`
5. **##7 Segment Display**
6. `set_property -dict { PACKAGE_PIN W7 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[0]}]`
7. `set_property -dict { PACKAGE_PIN W6 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[1]}]`
8. `set_property -dict { PACKAGE_PIN U8 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[2]}]`
9. `set_property -dict { PACKAGE_PIN V8 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[3]}]`
10. `set_property -dict { PACKAGE_PIN U5 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[4]}]`
11. `set_property -dict { PACKAGE_PIN V5 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[5]}]`
12. `set_property -dict { PACKAGE_PIN U7 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[6]}]`
13. `set_property -dict { PACKAGE_PIN V7 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[7]}]`
14. `set_property -dict { PACKAGE_PIN U2 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[0]}]`
15. `set_property -dict { PACKAGE_PIN U4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[1]}]`
16. `set_property -dict { PACKAGE_PIN V4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[2]}]`
17. `set_property -dict { PACKAGE_PIN W4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[3]}]`
18. **##Pmod Header JC**
19. `set_property -dict { PACKAGE_PIN K17 IOSTANDARD LVCMOS33 } [get_ports {dht11_data_0}];#Sch name = JC1`
20. **##USB-RS232 Interface**
21. `set_property -dict { PACKAGE_PIN B18 IOSTANDARD LVCMOS33 } [get_ports usb_uart_rxd]`
22. `set_property -dict { PACKAGE_PIN A18 IOSTANDARD LVCMOS33 } [get_ports usb_uart_txd]`
23. **## Configuration options, can be used for all designs**
24. `set_property CONFIG_VOLTAGE 3.3 [current_design]`
25. `set_property CFGBVS VCCO [current_design]`
26. **## SPI configuration mode options for QSPI boot, can be used for all designs**
27. `set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]`
28. `set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]`
29. `set_property CONFIG_MODE SPIx4 [current_design]`
30. **## Configuration options, can be used for all designs**
31. `set_property CONFIG_VOLTAGE 3.3 [current_design]`
32. `set_property CFGBVS VCCO [current_design]`
33. **## SPI configuration mode options for QSPI boot, can be used for all designs**
34. `set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]`
35. `set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]`
36. `set_property CONFIG_MODE SPIx4 [current_design]`

➤ AXI Connect ➔ AXI Template

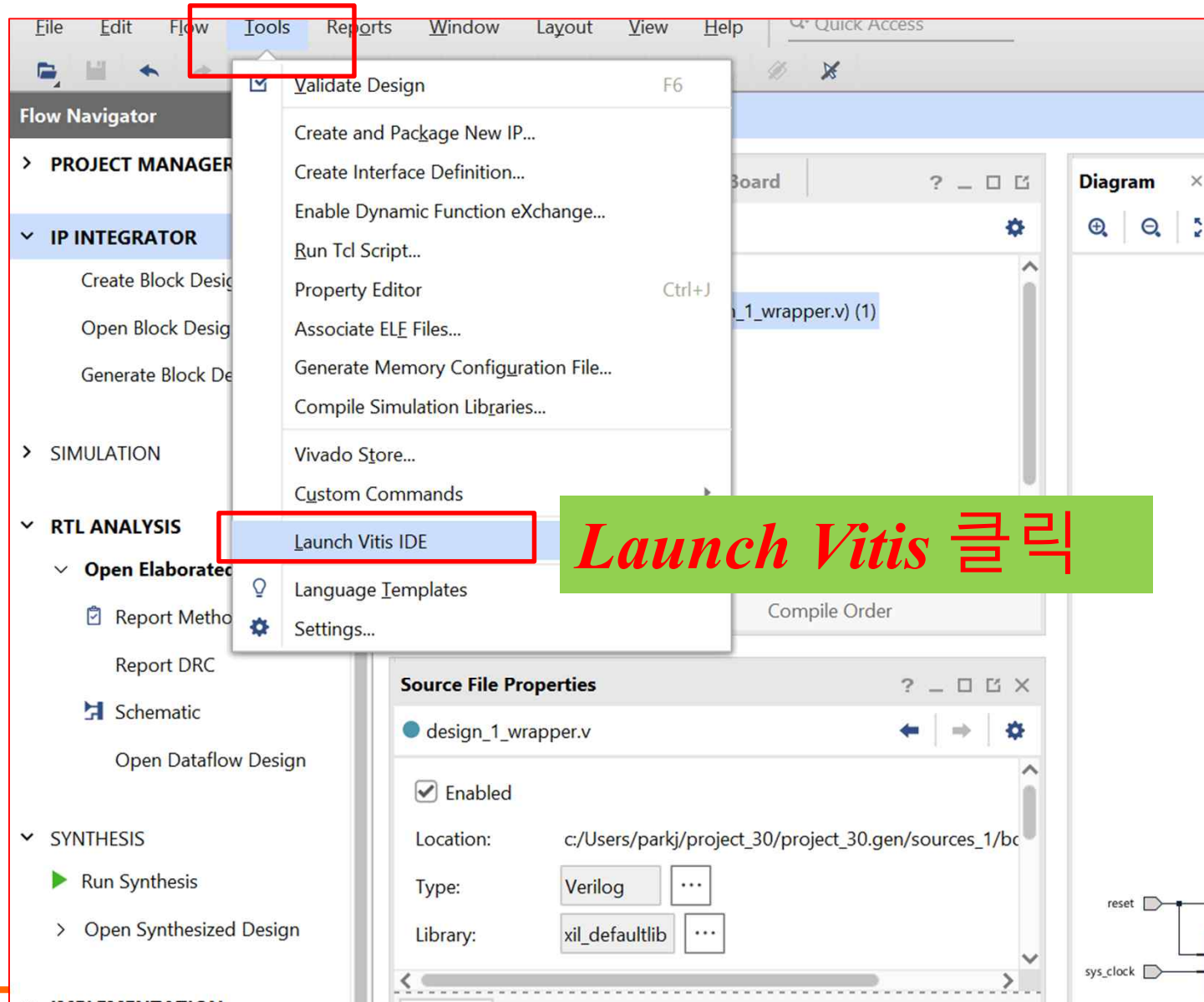
- *Create Block Design*
- *Blaze Uart & FND* 추가 및 연결
- *sys_clock & reset* 설정
- *FND, FNDsel, Dht11_data External*
- *XDC Constraints* 설정
- *Bitstream*
- *Export Hardware*

AMBA and AXI Connect

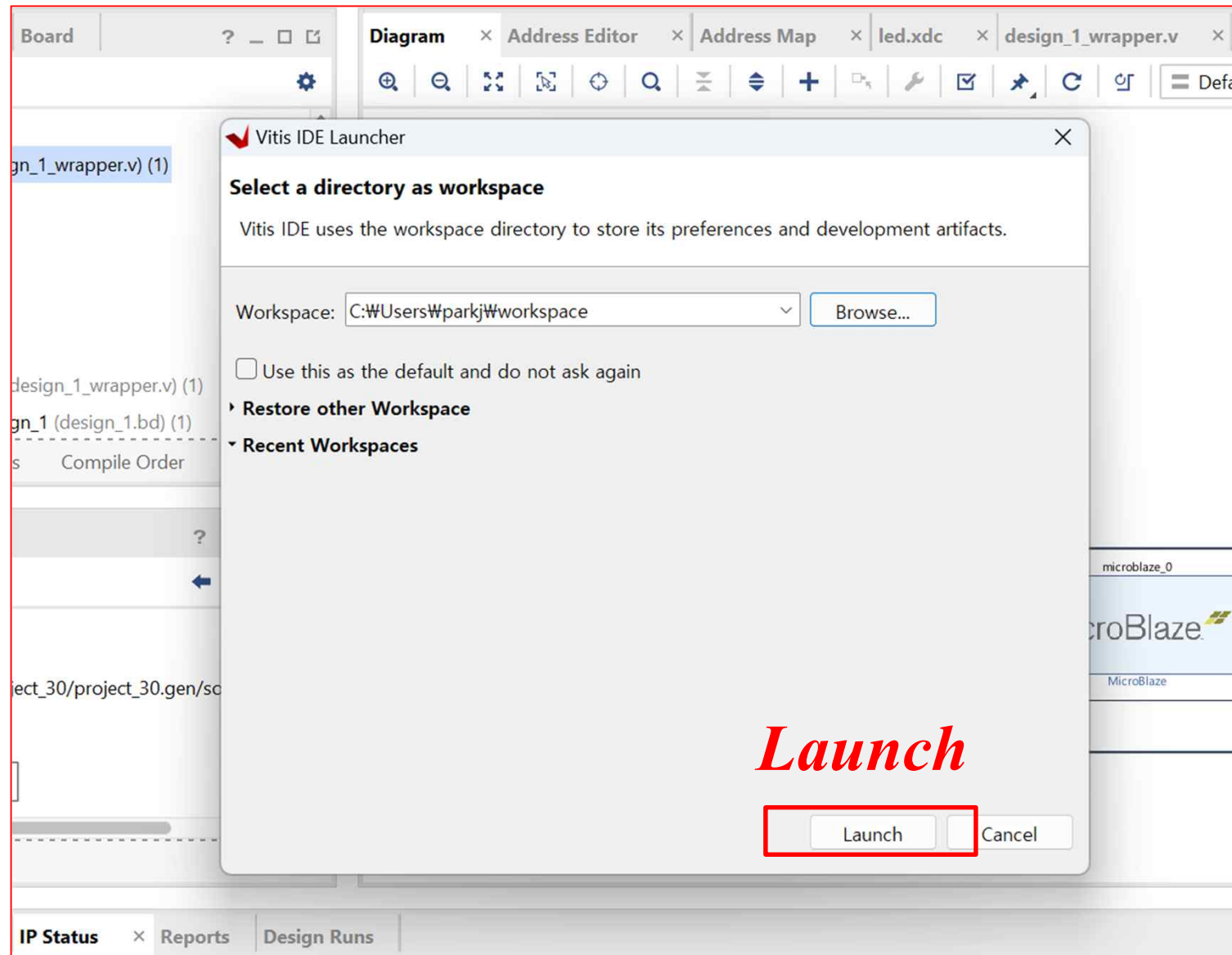
[Temperature–Humidity Sensor Module]

✓ *Application SW(Vitis)*

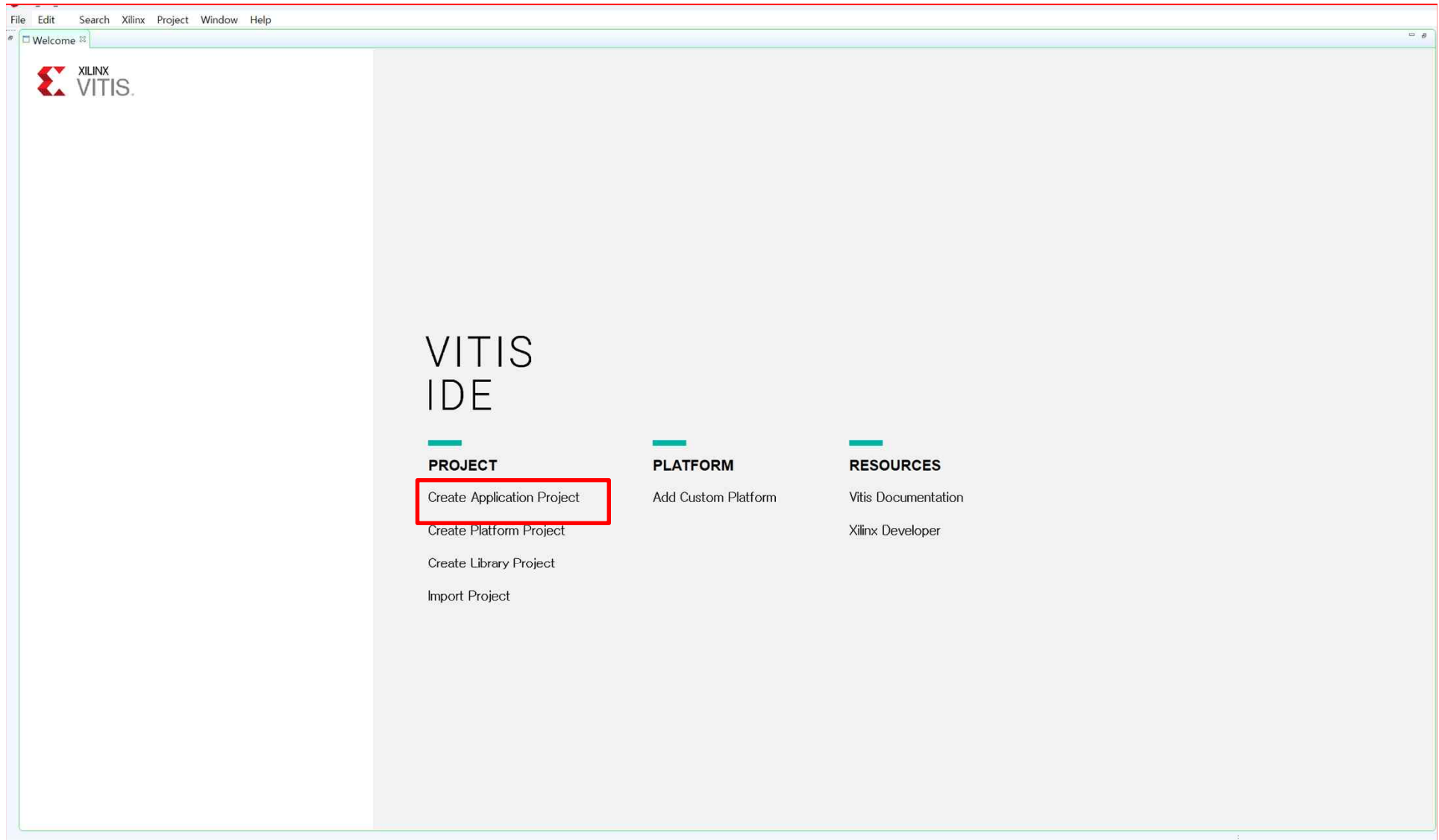
➤ AXI Connect ➔ AXI Template



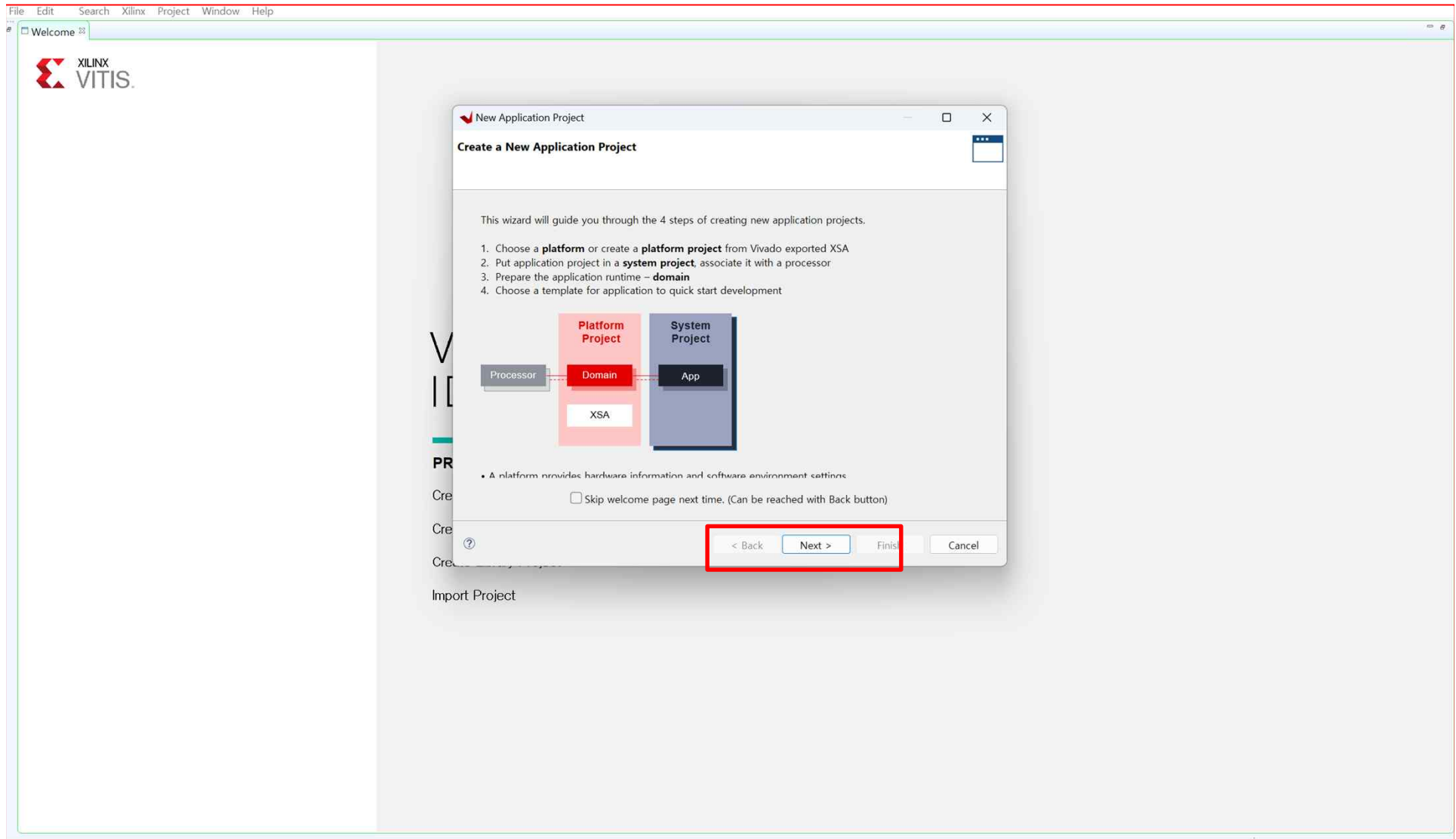
➤ AXI Connect ➔ AXI Template



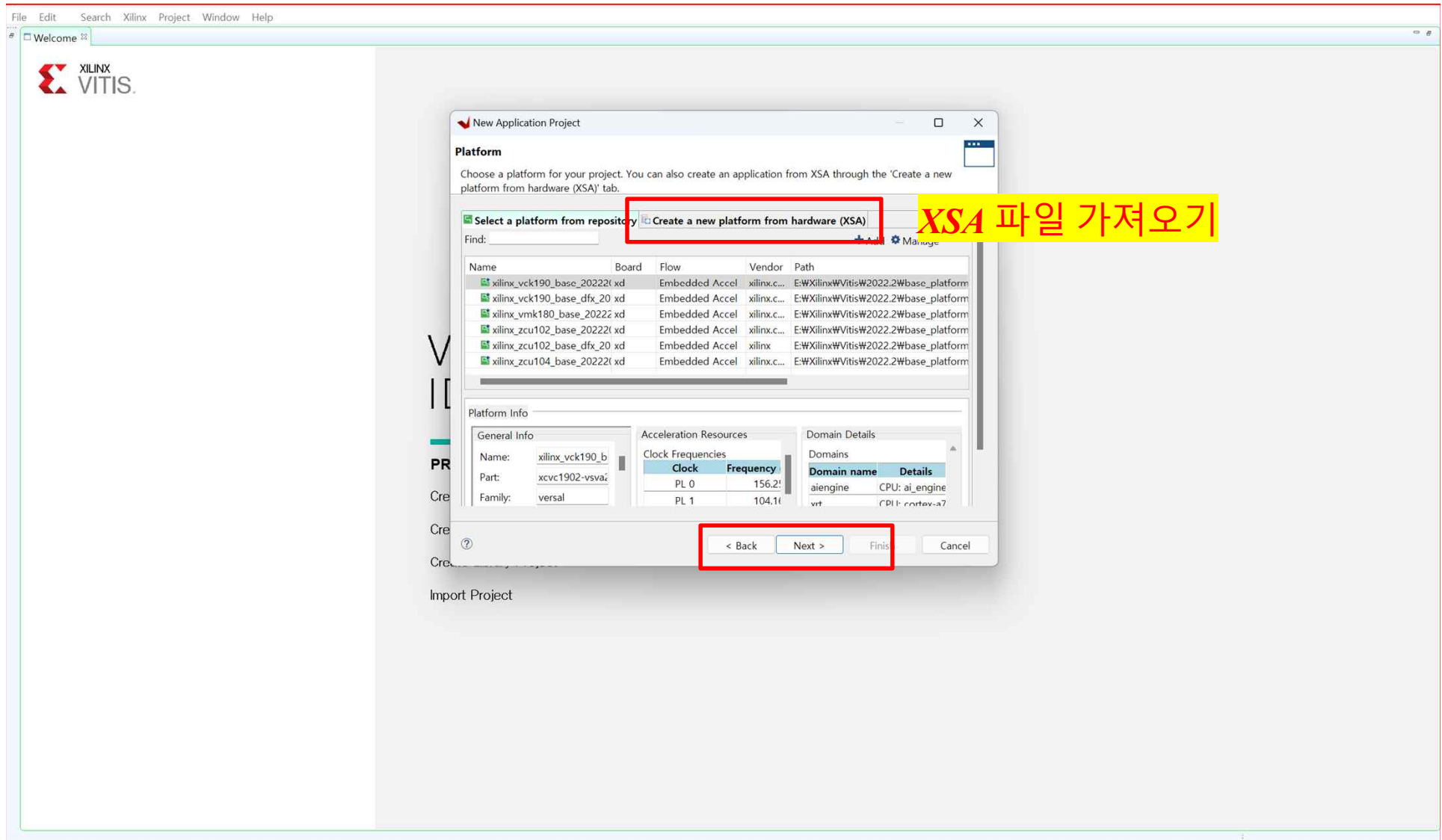
➤ AXI Connect ➔ AXI Template



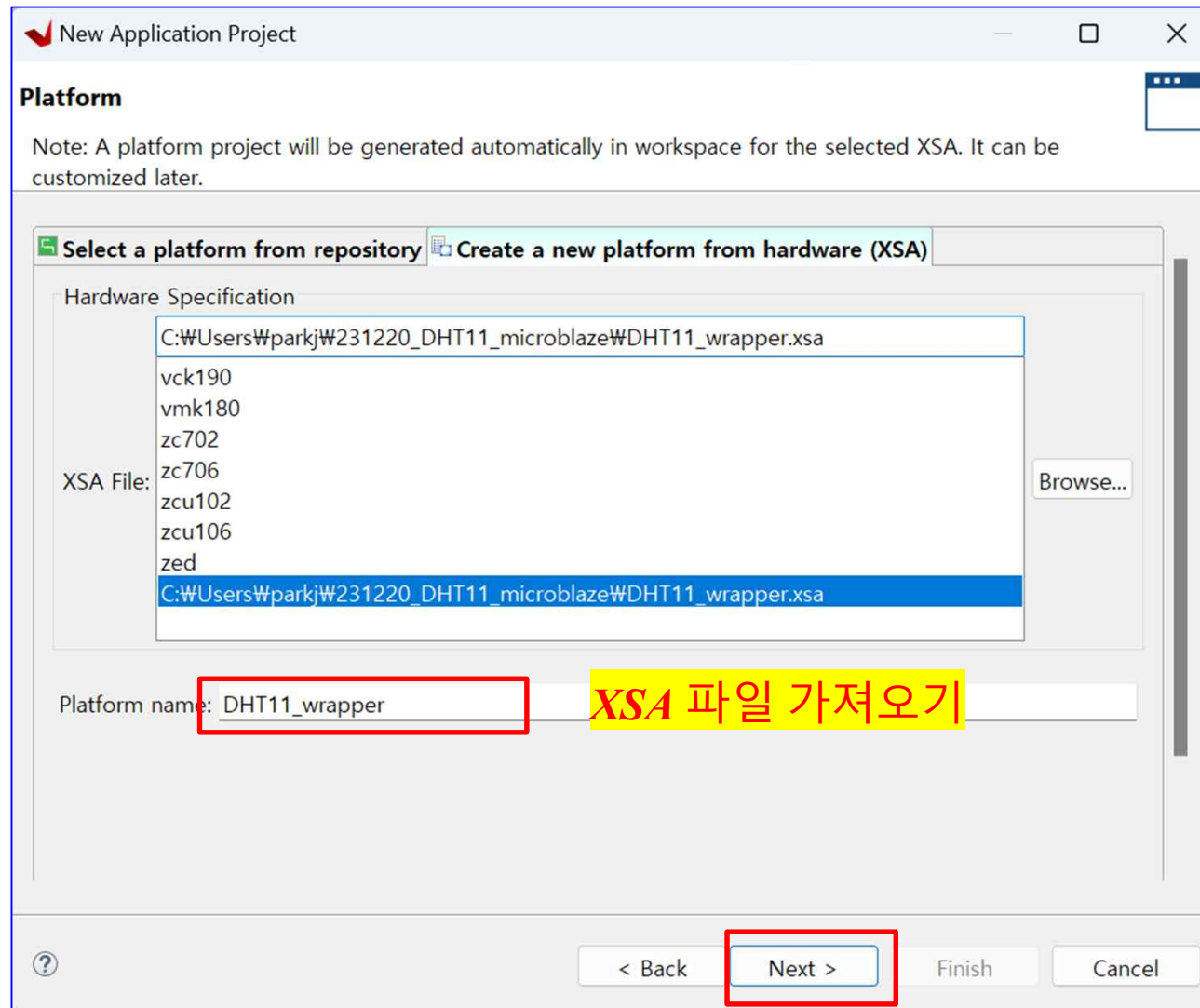
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

New Application Project

Application Project Details
Specify the application project name and its system project properties

Application project name:

System Project
Create a new system project for the application or select an existing one from the workspace ⓘ

Select a system project
+ Create new...

System project details

System project name:

Target processor
Select target processor for the Application project.

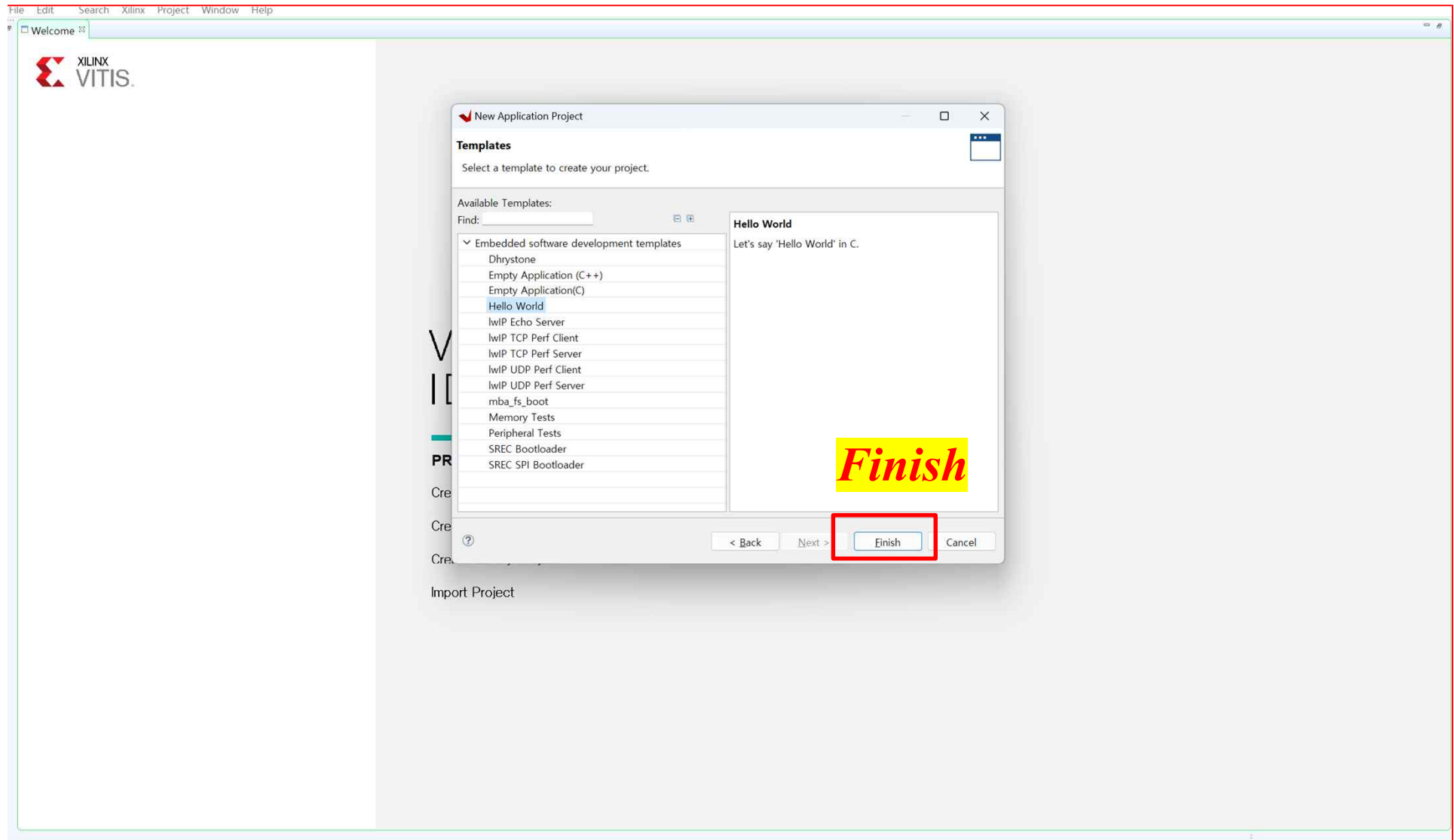
Processor	Associated applications
microblaze_0	DHT11

Show all processors in the hardware specification ☒ ⓘ

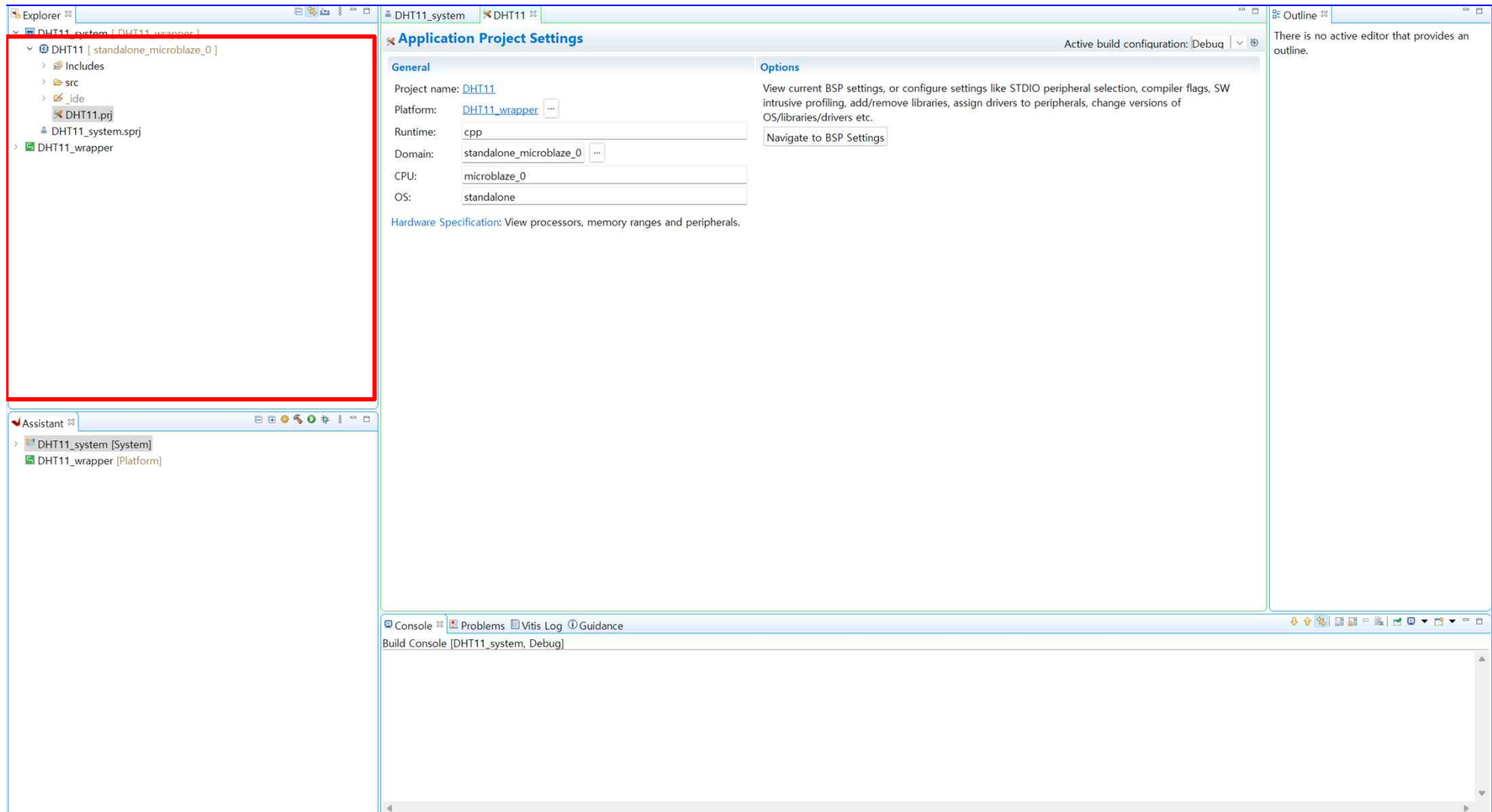
Next

? < Back **Next >** Finish Cancel

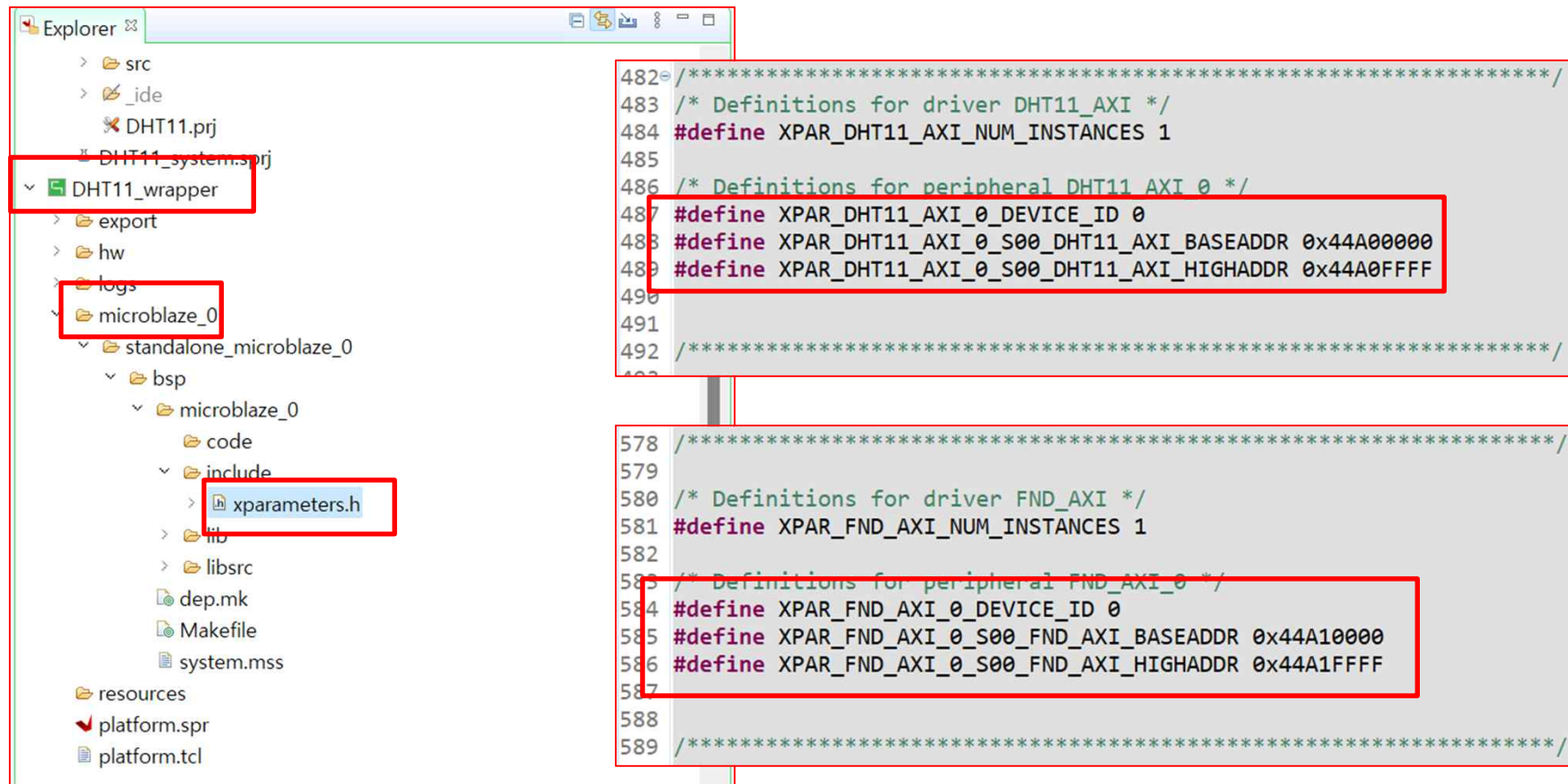
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



The image shows a project explorer on the left and a code editor on the right. In the project explorer, the following items are highlighted with red boxes:

- DHT11_wrapper** (under DHT11_system.spr)
- microblaze_0** (under standalone_microblaze_0)
- xparameters.h** (under include)

The code editor shows the following definitions:

```

482= /***** /
483 /* Definitions for driver DHT11_AXI */
484 #define XPAR_DHT11_AXI_NUM_INSTANCES 1
485
486 /* Definitions for peripheral DHT11_AXI_0 */
487 #define XPAR_DHT11_AXI_0_DEVICE_ID 0
488 #define XPAR_DHT11_AXI_0_S00_DHT11_AXI_BASEADDR 0x44A00000
489 #define XPAR_DHT11_AXI_0_S00_DHT11_AXI_HIGHADDR 0x44A0FFFF
490
491
492 /***** /
493
578 /***** /
579
580 /* Definitions for driver FND_AXI */
581 #define XPAR_FND_AXI_NUM_INSTANCES 1
582
583 /* Definitions for peripheral FND_AXI_0 */
584 #define XPAR_FND_AXI_0_DEVICE_ID 0
585 #define XPAR_FND_AXI_0_S00_FND_AXI_BASEADDR 0x44A10000
586 #define XPAR_FND_AXI_0_S00_FND_AXI_HIGHADDR 0x44A1FFFF
587
588
589 /***** /
    
```

BASEADDR 확인

➤ AXI Connect ➔ AXI Template

DiagramAddress EditorAddress MapDHT11.xdc

Assigned (5)

Unassigned (0)

Excluded (0)

Hide All

Name

Interface

Slave Segment

Master Base Address

Range

Master High Address

Network 0

/microblaze_0

/microblaze_0/Data (32 address bits : 4G)

/axi_uartlite_0/S_AXIS_AXIReg0x4060_000064K0x4060_FFFF

/DHT11_axi_0/S00_DHT11_AXIS00_DHT11_AXI_reg0x44A0_000064K0x44A0_FFFF

/fnd_axi_0/S00_fnd_AXIS00_fnd_AXI_reg0x44A1_000064K0x44A1_FFFF

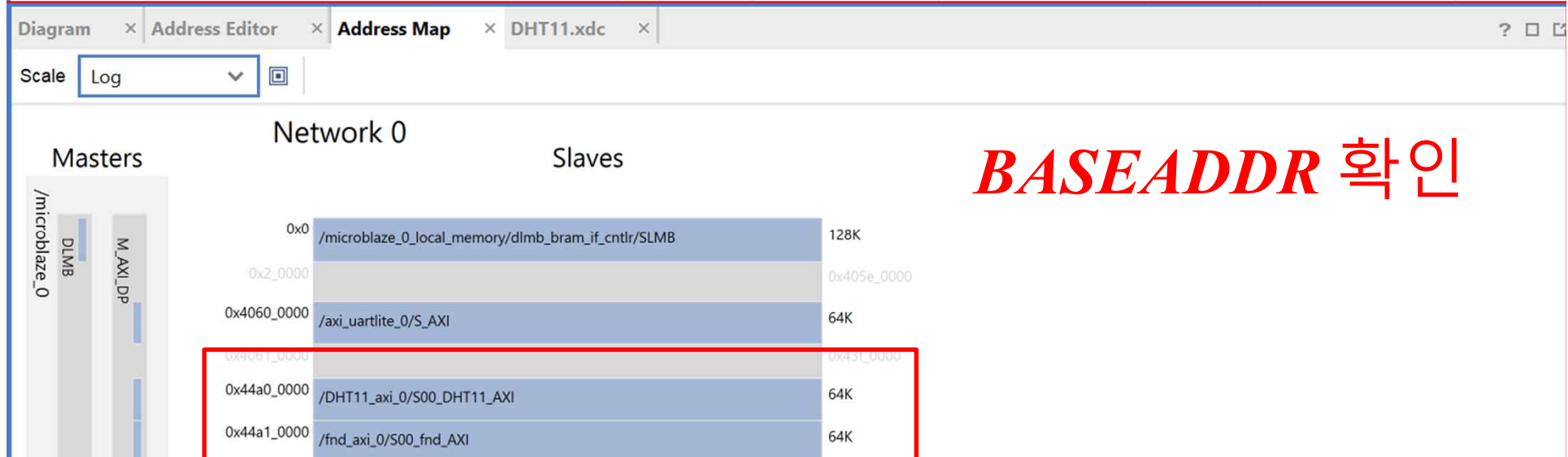
/microblaze_0_local_memory/dlmb_bram_if_cntlr/SLMBSLMBMem0x0000_0000128K0x0001_FFFF

Network 1

/microblaze_0

/microblaze_0/Instruction (32 address bits : 4G)

/microblaze_0_local_memory/ilmb_bram_if_cntlr/SLMBSLMBMem0x0000_0000128K0x0001_FFFF



BASEADDR 확인

➤ AXI Connect ➔ AXI Template

The screenshot shows the Xilinx IDE with the AXI Template code for helloworld.c. The code is highlighted with a red box. The text "Vitis Coding" is overlaid in red.

```
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29 * this Software without prior written authorization from Xilinx.
30 *
31 *
32 *
33 */
34 * helloworld.c: simple test application
35 *
36 * This application configures UART 16550 to baud rate 9600.
37 * PS7 UART (Zynq) is not initialized by this application, since
38 * bootrom/bsp configures it to baud rate 115200
39 *
40 *
41 * | UART TYPE   BAUD RATE |
42 * |-----|
43 * | uartns550   9600
44 * | uartrlite   Configurable only in HW design
45 * | ps7_uart    115200 (configured by bootrom/bsp)
46 */
47
48 #include <stdio.h>
49 #include "platform.h"
50 #include "xil_printf.h"
51
52
53 int main()
54 {
55     init_platform();
56
57     print("Hello World\n\r");
58     print("Successfully ran Hello World application");
59     cleanup_platform();
60     return 0;
61 }
62
```

Vitis Coding

➤ AXI Connect ➔ AXI Template

Helloworld.c

```
#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xparameters.h"
#include "sleep.h"
#include "xil_io.h"
```

주소 확인

```
#define AXI_FND_BASEADDR 0x44A10000
#define AXI_FND_DIGIT1 AXI_FND_BASEADDR + 0x00
#define AXI_FND_DIGIT2 AXI_FND_BASEADDR + 0x04
#define AXI_FND_DIGIT3 AXI_FND_BASEADDR + 0x08
#define AXI_FND_DIGIT4 AXI_FND_BASEADDR + 0x0c

#define AXI_DHT11_BASEADDR 0x44A00000
#define TEMPERATURE0 AXI_DHT11_BASEADDR + 0x00
#define TEMPERATURE10 AXI_DHT11_BASEADDR + 0x04
#define HUMYDITY0 AXI_DHT11_BASEADDR + 0x08
#define HUMYDITY10 AXI_DHT11_BASEADDR + 0x0c
```

```
int main()
{
    init_platform();

    print("Hello World\n\r");
    print("Successfully ran Hello World application");
}
```

Helloworld.c

```
u16 temp0 = 0;
u16 temp10 = 0;
u16 hum0 = 0;
u16 hum10 = 0;
```

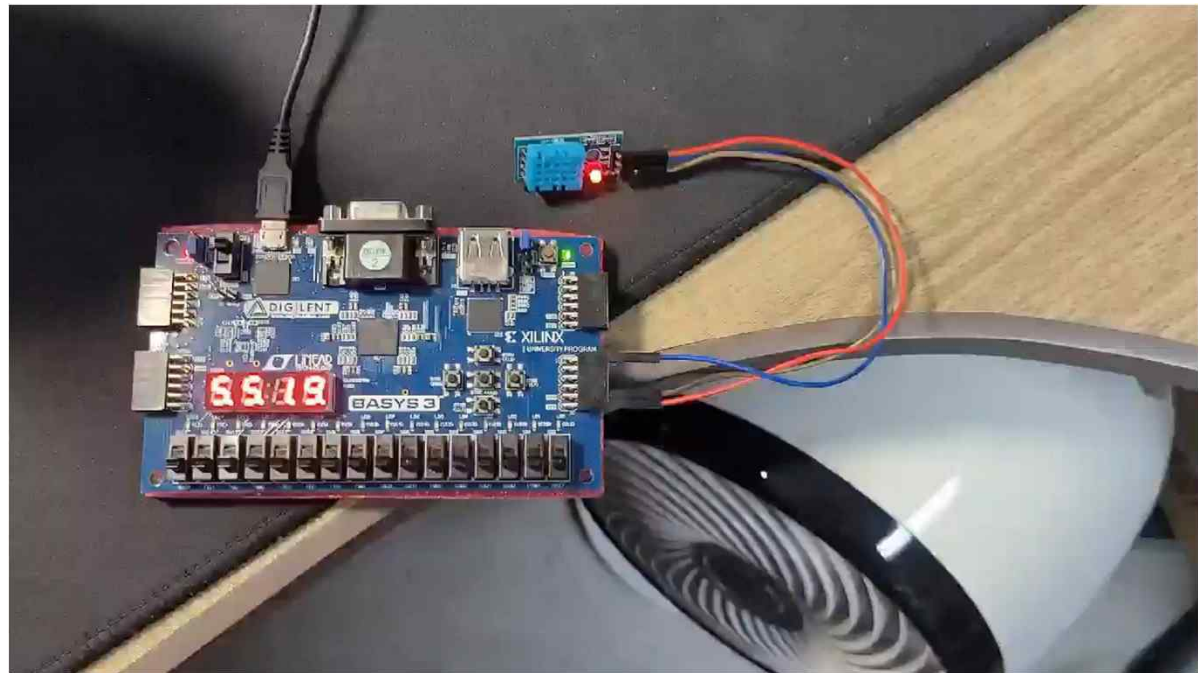
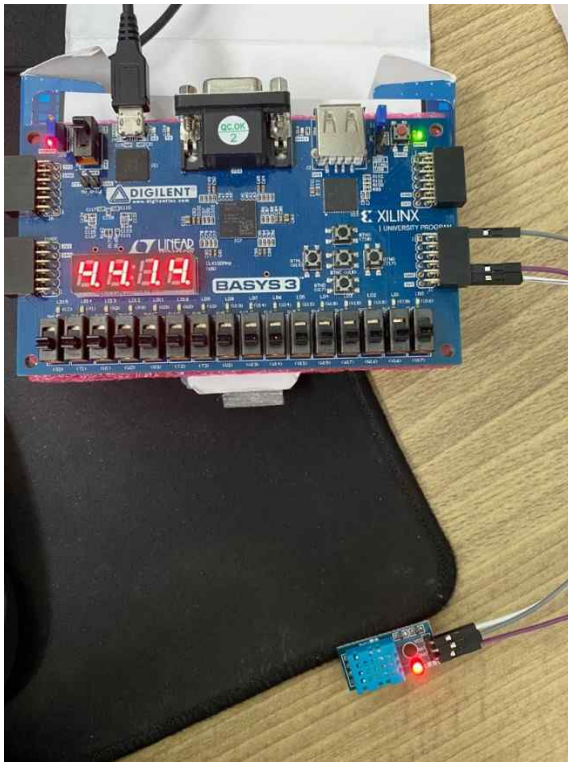
```
while(1)
{
    temp0=Xil_In32(TEMPERATURE0);
    temp10=Xil_In32(TEMPERATURE10);
    hum0=Xil_In32(HUMYDITY0);
    hum10=Xil_In32(HUMYDITY10);
```

```
Xil_Out32(AXI_FND_DIGIT1,temp0);
Xil_Out32(AXI_FND_DIGIT2,temp10);
Xil_Out32(AXI_FND_DIGIT3,hum0);
Xil_Out32(AXI_FND_DIGIT4,hum10);
```

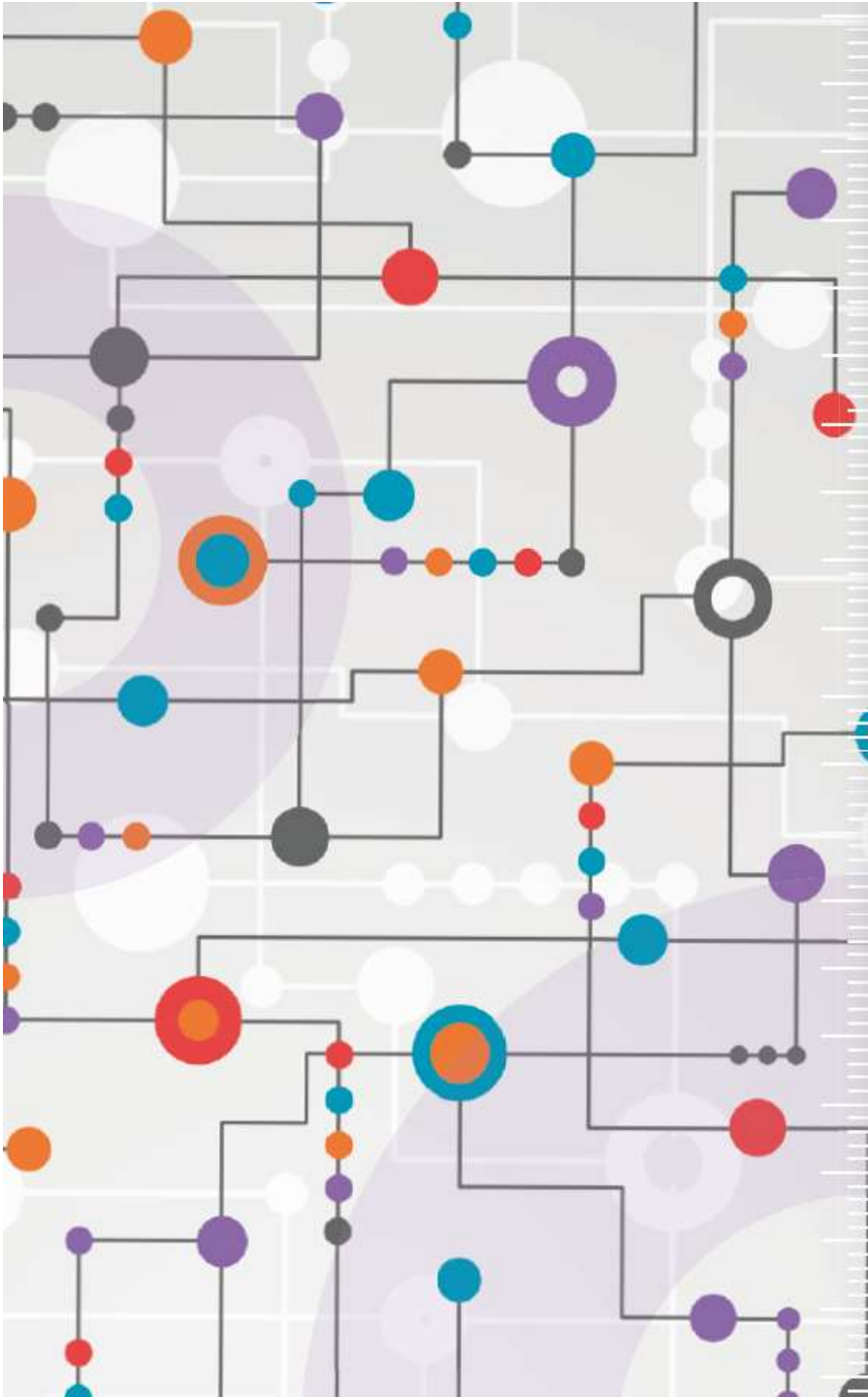
```
usleep(500000);
```

```
}
cleanup_platform();
return 0;
}
```


➤ Program Device & Check The Result



현재 온도, 습도가 출력되는 것을 확인



수고하셨습니다.

Inout Port Error

[*Reference*]

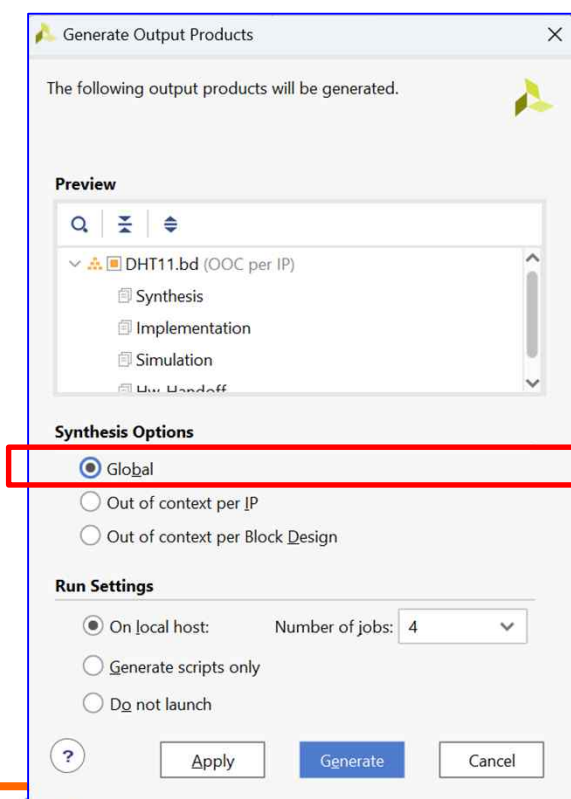
※ 오류 해결

➤ CustomIP에 *inout port* 사용시

➤ Inout 포트가 *output* 포트로만 적용됨. (*input x*)

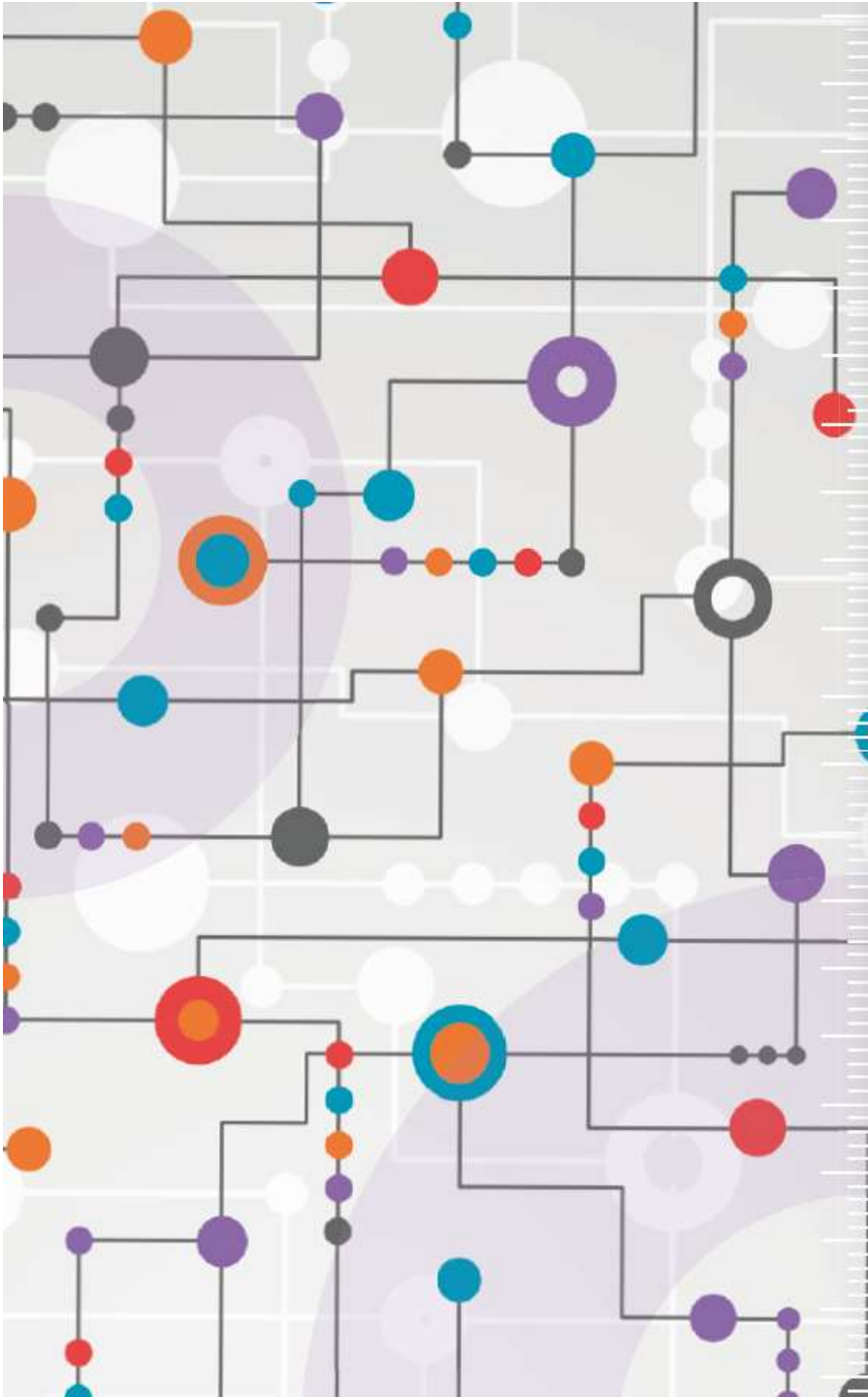
https://support.xilinx.com/s/question/0D54U00005ZMzumSAD/how-to-correctly-connect-packaged-ip-to-a-bidirectional-inout-port?language=en_US

diagram. This approach did NOT work when synthesized with the default values of "Generate Output Products"... by default that selects Synthesis Option "Out of context per IP". Using these defaults the top-level INOUT port/pin worked as an output only, and always read 0 for input. I had to change the Synthesis Option from the default to "Global" to get things working. No clue why, but that worked.



➤ 오류 해결을 위하여

Synthesis 옵션 *Global*로 수정



수고하셨습니다.